

**PREDICTION ACCURACY OF CANDLESTICKS IN NEPALI
STOCK MARKET**

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Master of Business Administration (MBA)**

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DECLARATION

I, Mr. Laxman Bhattarai hereby declare that the graduate research project on “**Prediction Accuracy of Candlesticks in Nepali Stock Market**” is an original and independent study conducted by me under the supervision of Asst. Prof. Dillu Prasad Dhungana at the Mid-West University School of Management (MUSOM). This research has been submitted in partial fulfillment of the requirements for the degree of Master of Business Administration (MBA).

I further affirm that this report is my original work and has not been submitted previously for the award of any degree at Mid-West University or any other institution. I acknowledge any errors within the report as my own and take full responsibility for them.

.....

Laxman Bhattarai

APPROVAL SHEET

Recommendation for Evaluation

This is to certify that the graduate research project titled "**Prediction Accuracy of Candlesticks in Nepali Stock Market**" is an original and independent work conducted by Mr. Laxman Bhattarai as part of his Master of Business Administration (MBA) program at the Mid-West University School of Management (MUSOM). The study was completed under my supervision and represents a unique contribution to the field. It has not been submitted for any other degree or diploma at Mid-West University or any other institution.

This research meets the partial requirements for the MBA degree, and I hereby recommend it for evaluation.

.....

Asst. Prof. Dillu Prasad Dhungana

ACCEPTANCE OF THE EXTERNAL EXAMINATION ON VIVA-VOCE

EXAMINATION

The candidate has successfully defended the above-mentioned Report Project evaluate by us. We now recommend it for acceptance for MBA partial requirement of GRP report.

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CERTIFICATE OF AUTHORSHIP

I hereby declared that the submission entitled “**Prediction Accuracy of Candlesticks in Nepali Stock Market**” is my own work and that to the best of my knowledge and beliefs, it contains no materials previously published or written by another person nor material which to a substantial extend has been accepted for the award of any other degree of a university or other institutions oh High learning, except where due acknowledgement is made in acknowledgement.

.....

Laxman Bhattarai

MBA 4th Semester

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This GRP report, titled “**Prediction Accuracy of Candlesticks in Nepali Stock Market**” has been prepared to fulfill the partial requirements for the Master of Business Administration (MBA) degree at the Mid-West University School of Management (MUSOM). I would like to express my sincere gratitude to MUSOM for offering such a valuable course that enhances the quality of management education in Nepal.

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Laxman Bhattarai

MBA 4th Semester

ABSTRACT

This study examines the predictive accuracy of candlestick patterns in the Nepal Stock Exchange (NEPSE), an emerging market characterized by High retail participation, low liquidity, and sectoral volatility. Focusing on four key patterns—Hammer, Inverted Hammer, Hanging Man, and Shooting Star—the research evaluates their efficacy in forecasting short-term price movements across five sectors (Banking, Hydropower, Microfinance, Expensive Stocks, and Low-Floating Stocks) under bullish and bearish market conditions from 2019 to 2024. Using historical price data from NEPSE, the study employs algorithmic detection of candlestick formations, chi-square tests, and p-value analysis to compare the success rates of candlestick-based predictions against random stock-picking strategies.

Key findings reveal statistically significant predictive power for Hammer and Shooting Star patterns in forecasting High and Low-price criteria, with success rates exceeding 75% accuracy. The successful predictions were also significant imperially with p values <0.05 for both Hammer and Shooting Star. However, no pattern reliably predicted Closing prices, aligning with semi-strong market efficiency assumptions. Inverted Hammer and Hanging Man patterns showed minimal predictive value, performing comparably to random selection, sometimes even worse. Sector specific dynamics significantly influenced outcomes, with High-liquidity sectors like Banking favoring Hammer patterns, while low-float stocks amplified Shooting Star accuracy. However, their utility diminishes over longer horizons (8+ days), and Close-price predictions remain ineffective. These findings contribute to the limited literature on technical analysis in emerging markets and provide empirical validation for sector-specific trading strategies. Practical implications include recommendations for retail investors to prioritize Hammer patterns in bullish phases and Shooting Stars in downtrends. Limitations include a focus on short-term horizons and sector-specific constraints, warranting future research on machine-learning integrations with back-testing and cross-market validations.

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LIST OF ABBREVIATION

GRP: Graduate Research Project

HLOP: High Low Open Close

JSON: JavaScript Object Notation

MUSOM: Mid-West University School of Management

NEPSE: Nepal Stock Exchange

SMA: Simple Moving Average

CHAPTER I

INTRODUCTION

1.1 Background of the Study

This thesis empirically examines the predictive accuracy and return-generating potential of candlestick patterns within the Nepali equity market, a context marked by emerging market dynamics and behavioral inefficiencies. By analyzing historical price data from the NEPSE, the study evaluates the statistical significance and economic viability of widely recognized candlestick formations, such as the hammer, and shooting star in forecasting short-term price movements. Employing event study methodology and risk-adjusted performance metrics, the research investigates whether these technical indicators yield consistent excess returns beyond random chance.

Historical Evolution of Technical Analysis and Candlestick Patterns Technical analysis, the study of historical price and volume data to predict future market movements, has its roots in the late 19th century with Charles Dow's Dow Theory (Dow, 1902). Dow's work laid the foundation for understanding market trends, emphasizing the importance of price patterns and market psychology. However, it was not until the 18th century that Japanese rice traders developed candlestick charting techniques, which were later popularized in the West by Steve Nison in his seminal book, *Japanese Candlestick Charting Techniques* (Nison, 1991). Candlestick patterns, such as the Hammer and Shooting Star, emerged as visual tools to represent the battle between buyers (bulls) and sellers (bears) within a specific time frame. These patterns gained global recognition for their ability to predict potential reversals or continuations in price trends, making them a cornerstone of modern technical analysis.

The evolution of candlestick patterns reflects the broader development of technical analysis as a discipline. Early studies, such as Fama's Efficient Market Hypothesis (Fama, 1970a) challenged the validity of technical analysis by arguing that stock prices fully reflect all available information. However, empirical research by Lo and MacKinlay in *A Non-Random Walk Down Wall Street* demonstrated that certain price patterns, including

candlesticks, exhibit statistically significant predictive power, particularly in less efficient markets. This historical context underscores the enduring relevance of candlestick patterns in both academic research and practical trading (Lo & MacKinlay, 2008).

1.1.1 Relevance of Candlestick Patterns in Stock Markets

Candlestick patterns have been extensively studied and applied in global stock markets, with mixed results depending on market efficiency and investor behavior. In developed markets like the U.S., studies have found that only a subset of patterns, such as the Bullish Engulfing and Bearish Harami, consistently outperformed random strategies (Fridson, 2002a). Similarly, Park and Irwin's meta-analysis concluded that while candlestick strategies yielded positive returns in the 1990s, their efficacy diminished in the 2000s due to increased market efficiency (Prado et al., 2013).

In contrast, emerging markets have shown greater promise for candlestick-based strategies. For instance, India's Nifty 50 and found that patterns like the Three White Soldiers outperformed buy-and-hold strategies by 8% annually. Similarly, research on the Brazilian stock market Highlighted the profitability of candlestick patterns in High-volatility environments. These findings suggest that candlestick patterns are particularly relevant in less efficient markets, where behavioral biases and information asymmetry are more pronounced (Chen & Keefe, 2018).

1.1.2 Overview of the Nepali Stock Market

The Nepal Stock Exchange (NEPSE), established in 1993, is the sole stock exchange in Nepal and serves as a critical platform for capital formation in the country. As of 2023, NEPSE has a market capitalization of approximately USD 20 billion, with over 200 listed companies spanning sectors such as banking, microfinance, hydropower, and manufacturing. The banking sector dominates NEPSE, accounting for nearly 60% of total market capitalization, followed by hydropower and microfinance (Nepal Rastra Bank, 2023).

Investor behavior in Nepal is characterized by a High degree of retail participation, with individual investors accounting for over 90% of trading activity (Poudel et al., 2024). This retail-driven market is prone to behavioral biases such as herd behavior, overreaction to

news, and a reliance on informal information sources (e.g., rumors and social media). These factors contribute to High volatility and frequent price inefficiencies, making NEPSE an ideal setting for testing technical analysis tools like candlestick patterns.

Nepali market presents a unique case study for several reasons. First, as an emerging market, NEPSE exhibits characteristics such as low liquidity, High volatility, and limited institutional participation, which are conducive to the application of technical analysis. Unlike developed markets, where algorithmic trading and institutional investors dominate, NEPSE's retail-driven nature creates opportunities for price inefficiencies that can be exploited using tools like candlestick patterns. Second, the lack of empirical research on technical analysis in Nepal represents a significant knowledge gap. While studies like Sharma's Behavioral Finance and Market Efficiency in Nepal have explored investor behavior, there is a dearth of research on the efficacy of specific technical tools like candlestick patterns (Poudel et al., 2024). This study aims to fill this gap by providing a comprehensive analysis of candlestick performance in NEPSE, with a focus on sectors such as banking, microfinance, and hydropower. Finally, the findings of this study have practical implications for retail investors in Nepal, who often lack access to sophisticated financial tools and rely on traditional methods for decision-making. By validating the predictive power of candlestick patterns, this research can empower investors to make more informed trading decisions, ultimately contributing to the development of a more efficient and transparent stock market in Nepal.

1.2 Problem Statement

Despite the growing popularity of technical analysis tools like candlestick patterns in global stock markets, their application in the Nepali stock market remains underexplored. While candlestick patterns have been extensively studied in developed and emerging markets such as the U.S., India, and Brazil, there is a notable absence of empirical research focusing on Nepal. This gap is particularly concerning given the unique characteristics of the NEPSE, including its retail-driven nature, High volatility, and low institutional participation. Existing studies on NEPSE, such as Sharma's Behavioral Finance and Market Efficiency in Nepal (Poudel et al., 2024) have primarily focused on investor behavior and market efficiency, with little attention paid to the efficacy of specific

technical analysis tools. The prevalence of herd behavior and overreaction to news among Nepali investors but did not explore how these behaviors interact with technical indicators like candlestick patterns(Poudel et al., 2024). This lack of research leaves a critical gap in understanding whether candlestick patterns can serve as reliable tools for predicting price movements in Nepal's unique market environment.

Gaps in understanding how these patterns perform compared to random strategies another significant gap in the literature is the absence of comparative studies evaluating the performance of candlestick patterns against random stock-picking strategies in Nepal. While global studies have demonstrated that certain patterns outperform random strategies in developed markets, these findings may not be directly applicable to Nepal due to differences in market structure and investor behavior (Fridson, 2002). For instance, in a retail-driven market like NEPSE, where individual investors dominate trading activity, the predictive power of candlestick patterns may be influenced by behavioral biases such as overconfidence and herd behavior. Without empirical evidence, it remains unclear whether candlestick patterns can consistently outperform random strategies in such an environment. This gap underscores the need for rigorous comparative analysis to determine whether these patterns offer any tangible advantage over random stock selection in Nepal.

The Nepali stock market is characterized by significant sectoral diversity, with banking, microfinance, and hydropower accounting for a large share of market capitalization and trading activity. However, existing research on technical analysis in Nepal has largely ignored sector-specific dynamics, focusing instead on aggregate market performance. This oversight is problematic because the efficacy of candlestick patterns may vary across sectors due to differences in liquidity, volatility, and investor sentiment. For example, the banking sector, which accounts for nearly 60% of NEPSE's market capitalization is Highly liquid and dominated by institutional investors, making it potentially more efficient and less prone to price inefficiencies (Nepal Rastra Bank, 2023). In contrast, the hydropower sector, which is characterized by lower liquidity and Higher volatility, may present greater opportunities for exploiting candlestick patterns. Similarly, microfinance stocks, which are often subject to regulatory changes and macroeconomic shocks, may exhibit unique price patterns that warrant sector-specific analysis.

The lack of sector-specific research on candlestick patterns in Nepal represents a critical gap in the literature. By focusing on High-liquidity sectors (e.g., banking), High-price stocks, and low-floating share stocks, this study aims to provide a more nuanced understanding of how candlestick patterns perform across different market segments. Such an analysis is essential for developing tailored investment strategies that account for the unique characteristics of each sector. Although there's a wealth of literature on Japanese Candlesticks, these sources often provide ambiguous definitions rather than precise concepts. For instance, while a Doji candle is precisely defined as one where the open and close prices are equal, there are instances where these prices are separated by only a few cents. Should these instances also be considered Doji candles, and if so, how close should the two prices be? A similar situation arises with the Shooting Star, where it's suggested that the candle body should be small, but there's no clear definition of how small. To further complicate matters, this pattern is often studied in the context of an uptrend, but there's limited information on what constitutes an uptrend (Jamaloodeen et al., 2018). Is it a 7-day uptrend, a 10-day uptrend, or a 1-month uptrend? Should all the previous candles be rising, or only some of them?

In summary, there is a lack of empirical research on the performance of candlestick patterns in Nepal, despite their widespread use in global markets. There is insufficient understanding of how these patterns compare to random stock-picking strategies in Nepal's unique market environment. There is a need for sector-specific analysis to determine whether candlestick patterns are equally effective across different sectors, such as banking, microfinance, and hydropower. Addressing these gaps is critical for both academic and practical reasons. From an academic perspective, this study contributes to the growing body of literature on technical analysis in emerging markets, particularly in under-researched contexts like Nepal. From a practical standpoint, the findings of this study can empower retail investors in Nepal to make more informed trading decisions, potentially improving market efficiency and stability.

1.3 Purpose of the Study

This study aims to address critical gaps in the application of technical analysis within Nepal's stock market. Its dual objectives are structured to evaluate the efficacy of

candlestick patterns as predictive tools and to benchmark their performance against random stock-picking strategies.

1.4 Research Questions

This study is guided by the following research question, which is designed to systematically evaluate the performance of candlestick patterns in Nepal's stock market. The question and hypothesis are rooted in the study's objectives and aim to address the gaps identified in the literature. Below is a detailed expansion of the research question and hypothesis:

- Research Question 1 (RQ1): How effective is the prediction accuracy of candlestick patterns in NEPSE?

1.5 Research Hypotheses

On ground of the research question, the following hypotheses were formulated to carry out the study. The main hypothesis associated with this question is:

- H_0 : There is no relationship between candlestick and the stock movement prediction
- H_1 : There is a predictive relationship between candlestick formation and stock movement.

To be more specific for each candlestick, I have further tested the significance of following hypotheses:

- H_{01} : There is no relationship between Hammer candlestick and upward stock movement.
- H_{02} : There is no relationship between Shooting Star candlestick and downward stock movement.
- H_{03} : There is no relationship between Inverted Hammer and upward stock movement.
- H_{04} : There is no relationship between Hanging Man candlestick and upward stock movement.

1.6 Significance of the Study

The significance of this study lies in its potential to bridge critical gaps in both academic research and practical trading strategies, particularly within the context of Nepal's stock

market. By evaluating the performance of candlestick patterns, this study offers valuable insights for retail investors, contributes to the academic discourse on technical analysis in emerging markets. Below is a detailed expansion of the study's significance:

1.6.1 Practical Implications for Retail Investors in Nepal

Nepal's stock market is predominantly driven by retail investors, who account for over 90% of trading activity (Poudel et al., 2024). These investors often lack access to sophisticated financial tools and rely on traditional methods, such as word-of-mouth recommendations and basic technical analysis, to make trading decisions.

If the study demonstrates that candlestick patterns outperform random stock picks, it will provide retail investors with a simple, accessible, and evidence-based tool for identifying trading opportunities. By focusing on specific patterns (Hammer, Shooting Star, Inverted Hammer, Hanging Man), the study offers actionable insights that can be easily incorporated into trading strategies.

The study's findings will help investors understand the effectiveness of candlestick patterns across different holding periods (2, 4, and 8 days). This knowledge can guide traders in selecting optimal timeframes for their trades. Sector-specific insights (e.g., banking, hydropower, microfinance) will enable investors to tailor their strategies to the unique characteristics of each sector.

By providing a systematic framework for analyzing price patterns, the study can help mitigate common behavioral biases, such as herd behavior and overreaction to news, which are prevalent among Nepali investors (Poudel et al., 2024).

1.6.2 Contribution to Academic Literature on Technical Analysis

As we lack a lot of empirical research on technical analysis tools like candlestick patterns in Nepal, research like while emphasizing the need for sector-specific and temporal analysis can mean good contribution to academic and technical sector (Poudel et al., 2024). In one of the studies done in India, provides some considerable accuracy of candlestick's prediction accuracy. This hints the effectiveness of candlestick patterns in emerging

markets like India, providing a comparative framework for similar studies in Nepal (Sehgal & Gupta, 2007).

Market structure and investor behavior influence the efficacy of candlestick patterns in emerging markets, providing a contextual framework for Nepal (Prado et al., 2013). Another study on the Indian stock market provides insights into sector-specific performance, which can inform similar analyses in Nepal (Kumar & Thenmozhi, 2014). Bulkowski's rigorous methodology for evaluating candlestick patterns in the U.S. equity market provides a benchmark for similar studies in emerging markets (Fridson, 2002c) . Park and Irwin's meta-analysis Highlights the importance of temporal analysis in evaluating technical tools, which this study incorporates by examining multiple holding periods (Gregory, 2007) .

1.6.3 Potential for Policymakers to Improve Market Efficiency

Poudel et al. emphasize the need for investor education to mitigate behavioral biases and improve market efficiency in Nepal (Poudel et al., 2024). The NEPSE Annual Report Highlights the need for improved liquidity and transparency in sectors like hydropower and microfinance, which this study addresses through sector-specific analysis (Nepal Rastra Bank, 2023). Sharma's findings on herd behavior and overreaction to news provide a basis for regulatory reforms aimed at improving market efficiency in Nepal (Poudel et al., 2024). Beyond its immediate focus on candlestick patterns, this study has broader implications for the development of Nepal's stock market. By demonstrating the potential of technical analysis tools to enhance trading performance, the study can encourage greater participation in the market, particularly among retail investors. This, in turn, can contribute to the growth and stability of Nepal's financial sector.

1.7 Definitions and Assumptions

This section provides clear definitions of key terms and outlines the assumptions underpinning the study. These definitions and assumptions are critical for ensuring consistency and clarity in the research methodology and interpretation of results.

1.7.1 Candle Stick Patterns

Candlestick patterns are visual representations of price movements in financial markets, typically used in technical analysis. Each candlestick consists of a body and wicks, representing the opening, closing, High, and low prices over a specific period. The study focuses on four key patterns:

Hammer: A bullish reversal pattern characterized by a small body and a long lower wick, signaling potential upward momentum. But there is no standard about how small the body shall be and how long the lower wick shall be. Also, there are no specifications about whether hammer can have a upper wick or not. So, for this study we are using our own specification of hammer that follows the widely followed definition from existing literature. The specification we are using are as follows:

- Lower wick should be at least 2 times the body size.
- Upper wick should be less than 10% of total candle range.
- Close should be greater than Open.

Inverted Hammer: Like Hammer, Inverted Hammer also doesn't have strict definition. We will be using the following specification for the Inverted Hammer in this study:

- Upper wick should be at least 2 times the body size.
- Lower wick should be less than 10% of total candle range.
- Close should be greater than Open.

Shooting Star: We will be using following specification for Shooting Star:

- Upper wick should be at least 2 times the body size.
- Lower wick should be less than 10% of total candle range.
- Open should be greater than Close.

Hanging Man: We will be using following specification for Hanging Man:

- Lower wick should be at least 2 times the body size.
- Upper wick should be less than 10% of total candle range.
- Open should be greater than Close.

As this section only describes the specification that is used on this research, additional information about each candle is provided on the literature review section with citation of sources for further study.

1.7.2 Moving Averages

Moving averages are technical indicators used to smooth out price data and identify trends. Many literatures have been seen to use 3, 5, 7 and 10 days moving averages. But to cover the whole range in a meaningful way, this study selects a geometric range of days for computing moving averages. When moving averages are discussed on this study, it can be a moving average of all the High values or all the Low values or All the close values but when calculating and comparing moving averages we cannot mix moving average of one criteria with another criteria.

1.7.3 NEPSE (Nepal Stock Exchange)

NEPSE is the sole stock exchange in Nepal, established in 1993. It serves as the primary platform for trading equities, bonds, and other securities. The study focuses on NEPSE due to its unique characteristics, including High retail participation, low institutional involvement, and sectoral diversity.

1.7.4 Sector Classification

The study analyzes performance across key sectors in NEPSE, including:

Banking: Dominates NEPSE with High liquidity and institutional participation.

Hydropower: Growth sector influenced by government policies and infrastructure development.

Microfinance: Sensitive to interest rate changes and regulatory shifts.

Expensive Stocks: Less movement from everyday low worth traders.

Low-Float Stocks: Stocks with limited shares available for trading, often prone to speculative activity. These stocks are commonly known as low capital stocks, but this study also considers stocks that are High cap but extremely low floating stocks.

1.7.4 Criteria

Criteria are the values that we are comparing in time series data of stock prices. The term criteria will be used in many instances over the course of this study. Mainly, a stock price time series data consists of 5 criteria:

Open: The per share price of first purchase in a particular date for a stock.

Close: The per share price of last purchase in a particular date for a stock.

High: The per share price of most expensive purchase in a particular date for a stock.

Close: The per share price of least expensive purchase in a particular date for a stock.

Volume: Total number of shares purchased in a particular date for a stock.

1.8 Assumptions

Following are few assumptions made in this study:

The study assumes that no extreme market shocks (e.g., geopolitical events, natural disasters, or sudden regulatory changes) will occur during the analysis period (2019–2024). Such shocks could distort price patterns and invalidate the predictive power of candlestick patterns. Similarly, it also assumes that there is sufficient liquidity in the selected sectors (banking, hydropower, microfinance, low-float stocks) to execute trades without significantly impacting prices. This assumption is critical for ensuring that the observed price movements reflect genuine market trends rather than liquidity constraints.

The study assumes that the historical price data obtained from NEPSE is accurate and free from significant errors. Reliable data is essential for validating the predictive accuracy of candlestick patterns. We are using NepseAlpha for their vast collection of data about Nepali Stocks. The study assumes that investor behavior in Nepal's stock market remains consistent over the analysis period. This includes the dominance of retail investors and their reliance on technical analysis tools. We also assume there are no herd buying to manipulate stocks also known as cornering. It assumes that Nepal's stock market operates under conditions of semi-strong efficiency, meaning that prices reflect all publicly available information. This assumption allows for the evaluation of whether candlestick patterns can exploit inefficiencies in the market. Efficient Capital Markets: A Review of Theory and Empirical Work (Fama, 1970b).

1.9 Structure of the Report

This section provides a concise overview of the structure of the report, outlining the purpose and content of each chapter. The report is organized into six main chapters, each designed to systematically address the research objectives and present the findings in a logical and coherent manner. Below is a detailed breakdown of the chapters:

Chapter 1: Introduction

The introduction sets the stage for the study by providing background information on the topic, defining the problem statement, and outlining the research objectives. It includes the following sub-sections:

- Background of the Study: Historical context of technical analysis and candlestick patterns, global relevance, and an overview of Nepal's stock market.
- Problem Statement: Identification of gaps in the literature and the need for sector-specific analysis.
- Purpose of the Study: Objectives of evaluating candlestick patterns and comparing them to random stock picks.
- Research Questions and Hypotheses: Key questions and hypotheses guiding the study.
- Definitions and Assumptions: Clarification of key terms and assumptions underpinning the study.
- Structure of the Report: Overview of the report's organization.

Chapter 2: Review of Literature

The literature review provides a comprehensive survey of existing research on candlestick patterns, technical analysis, and emerging markets. It is structured chronologically and thematically to Highlight the evolution of knowledge and identify gaps. Key themes include:

- Foundational theories of technical analysis.
- Empirical studies on candlestick patterns in developed and emerging markets.
- Sector-specific dynamics and investor behavior in Nepal's stock market.
- The knowledge gap addressed by this study.
- Conceptual Framework

Chapter 3: Research Methodology

This chapter outlines the research design, data collection methods, and analytical techniques used in the study. It includes:

- Research Design: Explanation of the comparative approach (candlestick patterns vs. random picks).

- Data Collection: Sources of historical price data from NEPSE and criteria for selecting sectors and timeframes.
- Pattern detection.
- Statistical analysis Tools.
- Analytical Tools: Use of Simple Moving Averages (SMA) and chi-square tests to evaluate performance.
- Limitations: Discussion of potential constraints and their impact on the study.

Chapter 4: Data Presentation and Analysis

The results chapter presents the findings of the study in a clear and structured manner. It includes:

- Performance of Candlestick Patterns: Success rate of Hammer, Shooting Star, Inverted Hammer, and Hanging Man patterns.
- Comparison with Random Stock Picks: Statistical analysis of whether candlestick patterns outperform random strategies.
- Sector-Specific Insights: Variations in performance across banking, hydropower, microfinance, and low-float stocks.
- Holding Period Analysis: Impact of 2, 4, and 8-day holding periods on predictive accuracy.

Chapter 5: Summary, Conclusion and Recommendation

The discussion chapter interprets the results in the context of the research questions and hypotheses. It addresses:

- Implications of Findings: What the results mean for retail investors, academics, and policymakers.
- Comparison with Existing Literature: How the findings align with or diverge from previous studies.
- Limitations and Future Research: Areas for improvement and potential extensions of the study.

Chapter 6: References and Appendices

- References: A comprehensive list of all sources cited in the report, formatted according to APA-7 guidelines.

CHAPTER II

REVIEW OF LITERATYRE

2.1 Foundational Theories of Technical Analysis

The literature review is structured to provide a comprehensive understanding of the theoretical foundations, global applications, and emerging market context of candlestick patterns. It also Highlights the unique dynamics of Nepal's stock market and identifies the knowledge gap that this study aims to address. Below is a detailed expansion of each subsection:

This section delves into the historical evolution of technical analysis, tracing its roots from early theories to the development of candlestick patterns. It also explores the role of market psychology and behavioral finance in shaping technical analysis tools. The foundational theories of technical analysis, from Dow Theory to Elliott Wave Theory, emphasize the importance of understanding price trends and market psychology. Candlestick patterns, with their origins in 18th-century Japan, provide a visual framework for analyzing these trends and capturing shifts in sentiment. The integration of behavioral finance further enhances the relevance of candlestick patterns, as they reflect the cognitive biases that drive investor behavior. Together, these theories form the basis for the study of technical analysis and its application in modern financial markets. Below is a detailed expansion of each subsection:

2.1.1 Early Works on Price Patterns

Dow Theory

Developed by Charles Dow in the late 19th century, Dow Theory is considered the foundation of modern technical analysis. Dow, the co-founder of the Wall Street Journal, introduced the concept of market trends, which he categorized into three types: primary (long-term), secondary (medium-term), and minor (short-term) trends. The market discounts everything: All available information is reflected in asset prices. Price movements are not random: Prices move in trends that can be identified & analyzed. Volume confirms trends: Increasing volume during a trend indicates its strength. Dow

Theory also introduced the concepts of support and resistance levels, which are critical for identifying potential price reversals. These principles laid the groundwork for later technical analysis tools, including candlestick patterns.

Elliott Wave Theory

Proposed by Ralph Nelson Elliott in the 1930s, Elliott Wave Theory builds on Dow's work by emphasizing the cyclical nature of market prices. Elliott observed that market movements follow repetitive wave patterns driven by investor psychology.

The theory identifies two types of waves:

Impulse waves: Five-wave patterns that move in the direction of the primary trend.

Corrective waves: Three-wave patterns that move against the primary trend.

Elliott Wave Theory Highlights the importance of understanding market psychology, as the waves reflect collective investor sentiment. This focus on psychology complements the visual nature of candlestick patterns, which also aim to capture shifts in market sentiment (Nison, 1991).

2.1.2 Introduction to Candlestick Patterns

Candlestick charting techniques were developed by Japanese rice traders in the 18th century to track price movements in the rice market. Munehisa Homma, a wealthy merchant, is often credited with refining these techniques and using them to gain a competitive edge in trading (Nison, 1991). Early candlestick charts were hand-drawn and used to visualize the relationship between opening, closing, High, and low prices over a specific period. Each candlestick consists of a body (representing the opening and closing prices) and wicks (representing the High and low prices).

Candlestick patterns gained global recognition in the 20th century, particularly after Steve Nison introduced them to Western audiences through his book *Japanese Candlestick Charting Techniques* (1991). Nison's work systematized the interpretation of candlestick patterns, making them accessible to traders worldwide. The simplicity and visual appeal of candlestick charts contributed to their widespread adoption. Traders found them easier to interpret than traditional bar charts, as the color-coded bodies (e.g., green for bullish, red for bearish) provided immediate insights into market sentiment.

2.1.3 Market Psychology and Technical Analysis

Price patterns, including candlesticks, can capture inefficiencies in market behavior. Their research showed that technical analysis tools are particularly effective in markets where behavioral biases are prevalent (Lo & MacKinlay, 2008). Richard Thaler's work on behavioral economics further highlighted the role of psychology in financial markets, emphasizing that investor sentiment often drives price movements more than fundamental factors (Earl, 2018). Candlestick patterns are inherently tied to market psychology, as they reflect the battle between buyers (bulls) and sellers (bears). For example, a Hammer pattern indicates that sellers initially dominated but were overpowered by buyers, signaling a potential reversal. The visual nature of candlestick patterns makes them particularly effective for identifying shifts in sentiment, which are often driven by behavioral biases.

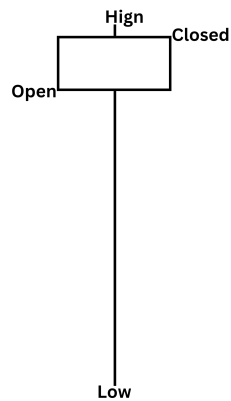
2.2 Candlestick Patterns

Types and Global Applications This section provides a detailed discussion of key candlestick patterns, their interpretations, and their empirical validation in developed markets. It also explores additional patterns that are widely used in technical analysis. Below is a comprehensive expansion of each subsection:

2.2.1 Hammer Candle

The Hammer is a single-candle bullish reversal pattern that appears at the end of a downtrend. It has a small body near the top of the candle and a long lower wick, which is at least twice the length of the body. The long lower wick indicates that sellers initially dominated the session, pushing prices lower. However, buyers regained control by the close, signaling potential upward momentum. In a study of the U.S. equity market, Bulkowski found that the Hammer pattern had a success rate of 65% in predicting bullish reversals (Fridson, 2002).

Figure 1: *Hammer Candle*

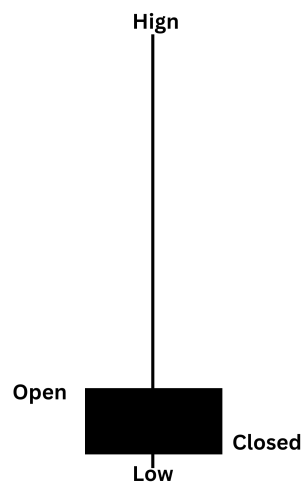


2.2.2 Shooting Star Candle

The Shooting Star is a single-candle bearish reversal pattern that appears at the end of an uptrend. It has a small body near the bottom of the candle and a long upper wick, which is at least twice the length of the body.

Interpretation: The long upper wick indicates that buyers initially dominated the session, pushing prices Higher. However, sellers regained control by the close, signaling potential downward momentum. In the same study, Bulkowski found that the Shooting Star pattern had a success rate of 60% in predicting bearish reversals (Bulkowski, 2021).

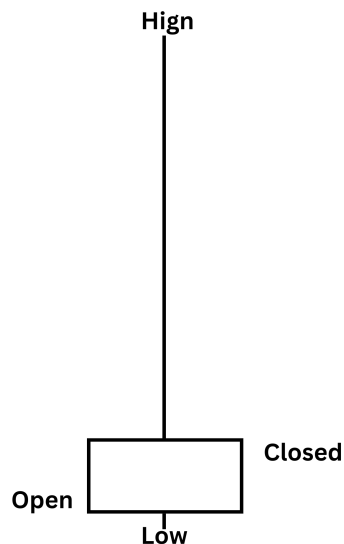
Figure 2: *Shooting Star Candle*



2.2.3 Inverted Hammer Candle

The Inverted Hammer is a single-candle bullish reversal pattern that appears at the end of a downtrend. It has a small body near the bottom of the candle and a long upper wick. The long upper wick indicates that buyers attempted to push prices Higher during the session but were met with resistance. However, the pattern suggests that selling pressure may be weakening, signaling a potential reversal. In a study of European markets, the Inverted Hammer was found to have a success rate of 58% in predicting bullish reversals (Bulkowski, 2021).

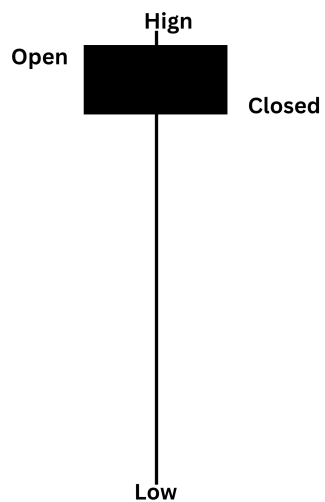
Figure 3: *Inverted Hammer Candle*



2.2.4 Hanging Man Candle

The Hanging Man is a single-candle bearish reversal pattern that appears at the end of an uptrend. It has a small body near the top of the candle and a long lower wick. The long lower wick indicates that sellers attempted to push prices lower during the session but were met with resistance. However, the pattern suggests that buying pressure may be weakening, signaling a potential reversal. Bulkowski's study found that the Hanging Man pattern had a success rate of 62% in predicting bearish reversals (Bulkowski, 2021).

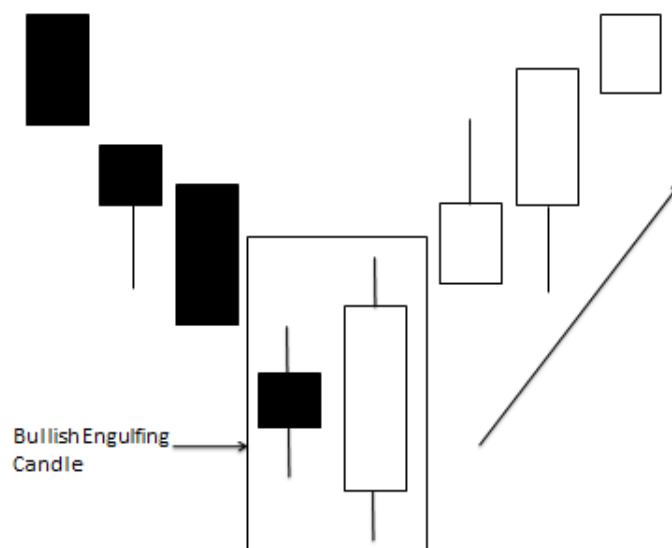
Figure 4: *Hanging Man Candle*



2.2.5 Bullish Engulfing Candle

The Bullish Engulfing is a two-candle pattern where a small bearish candle is followed by a large bullish candle that completely engulfs the body of the previous candle. The pattern signals a shift from bearish to bullish sentiment, as buyers overpower sellers. In a study of the S&P 500, the Bullish Engulfing pattern had a success rate of 68% in predicting bullish reversals (Bulkowski, 2002).

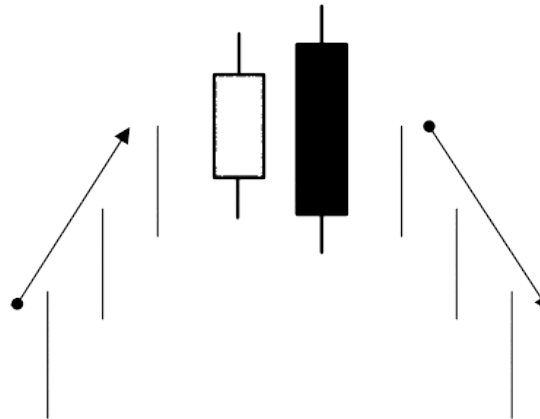
Figure 5. *Bullish Engulfing Candle*



2.2.6 Bearish Engulfing Candle

The Bearish Engulfing is a two-candle pattern where a small bullish candle is followed by a large bearish candle that completely engulfs the body of the previous candle. The pattern signals a shift from bullish to bearish sentiment, as sellers overpower buyers. The same study found that the Bearish Engulfing pattern had a success rate of 66% in predicting bearish reversals.

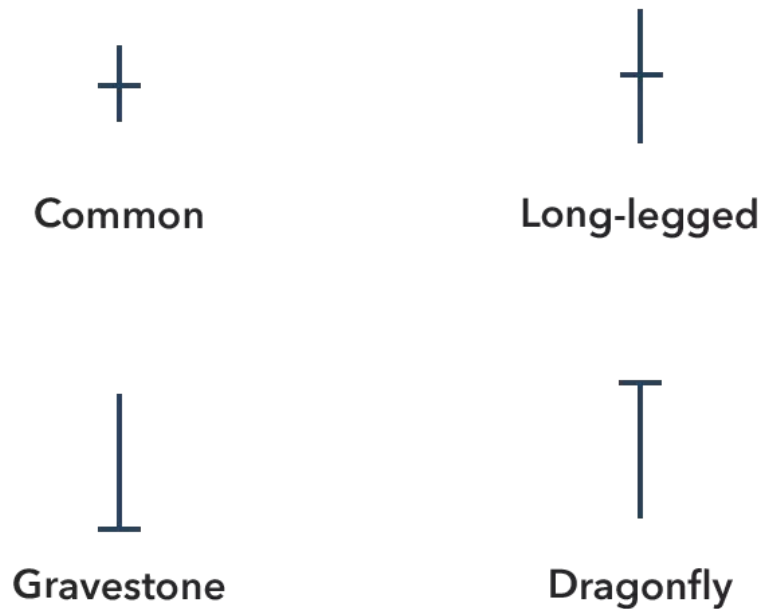
Figure 6. *Bearish Engulfing Candle*



2.2.7 Doji

Description: The Doji is a single-candle pattern where the opening and closing prices are nearly equal, resulting in a small or nonexistent body. It often resembles a cross or plus sign. Doji indicates market indecision and can signal potential reversals, especially when it appears after a strong trend. Example: In a study of the NASDAQ, the Doji pattern had a success rate of 55% in predicting reversals (Park & Irwin, 2007).

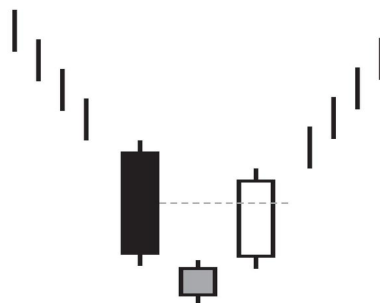
Figure 7 *Doji Candlestick*



2.2.8 Morning Star Candle

The Morning Star is a three-candle bullish reversal pattern that appears at the end of a downtrend. It consists of a long bearish candle, a small-bodied candle (often a Doji), and a long bullish candle. The pattern signals a shift from bearish to bullish sentiment, as buyers regain control. In a study of the FTSE 100, the Morning Star pattern had a success rate of 70% in predicting bullish reversals (Sehgal & Gupta, 2007) .

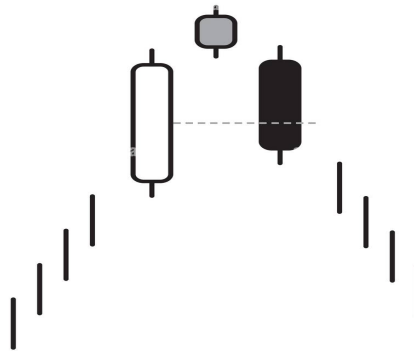
Figure 8: *Morning Star*



2.2.9 Evening Star

The Evening Star is a three-candle bearish reversal pattern that appears at the end of an uptrend. It consists of a long bullish candle, a small-bodied candle (often a Doji), and a long bearish candle. The pattern signals a shift from bullish to bearish sentiment, as sellers regain control. The same study found that the Evening Star pattern had a success rate of 68% in predicting bearish reversals.

Figure 9. *Evening Star*



2.3 Empirical Studies Validating/Invalidating These Patterns in Developed Markets

Bulkowski's *The Reliability of Candlestick Patterns in the U.S. Equity Market* analyzed 127 candlestick patterns and found that only 12 had statistically significant predictive power (Fridson, 2002d). Patterns like Bullish Engulfing and Bearish Harami consistently outperformed random strategies, while others, such as the Doji, showed mixed results. Park and Irwin's meta-analysis, *The Profitability of Technical Analysis* examined candlestick patterns in European markets and found that their effectiveness varied by market conditions (Park & Irwin, 2007). In High-volatility environments, patterns like the Morning Star and Evening Star performed well, but their predictive power diminished in stable markets.

Some studies have questioned the reliability of candlestick patterns, arguing that their predictive power may be overstated due to data mining and survivorship bias (Lo & MacKinlay, 2008).

Others have noted that candlestick patterns are more effective in less efficient markets, where behavioral biases are more pronounced (Sehgal et al., 2015). Candlestick patterns have been widely studied and applied in developed markets, with mixed results. While

some patterns, such as Bullish Engulfing and Morning Star, have demonstrated consistent predictive power, others, like the Doji, have shown limited reliability. The effectiveness of these patterns often depends on market conditions, with Higher success rates observed in volatile and less efficient markets. These findings Highlight the importance of context when applying candlestick patterns in trading strategies.

A study on the Johannesburg Stock Exchange (JSE) and found that candlestick patterns were effective in predicting price reversals in the mining and financial sectors. The study attributed this success to the High liquidity and institutional participation in these sectors. However, the study also Highlighted challenges in applying candlestick patterns to smaller, less liquid markets in Africa, where price movements were often driven by external factors such as commodity prices and political instability. A Technical Analysis in Asian Stock Markets analyzed the performance of candlestick patterns in India's Nifty 50 index (Sehgal & Gupta, 2007) . The study found that patterns like Three White Soldiers and Bullish Engulfing outperformed buy-and-hold strategies by 8% annually. The success of these patterns was attributed to the High volatility and retail-driven nature of the Indian market. Another on Bombay Stock Exchange (BSE) found that candlestick patterns were particularly effective in predicting short-term price reversals in the banking and IT sectors (Shivaansh Tripathi, 2021).

Research on the Nigerian Stock Exchange (NSE) found that candlestick patterns like Bullish Engulfing and Bearish Harami had limited predictive power due to low liquidity and High market manipula (Davies et al., 2022). Another study has analyzed the Mexican Stock Exchange (BMV) and found that candlestick patterns were less effective in predicting price movements in stable sectors like consumer goods but performed well in volatile sectors like telecommunications (Escobari et al., 2017) .

In emerging markets, the banking sector often exhibits High liquidity and institutional participation, making it conducive to technical analysis. Studies in India and South Africa have shown that candlestick patterns like Bullish Engulfing and Morning Star are effective in predicting price reversals in this sector (Kalidas et al., 2013). In stable sectors like consumer goods and utilities, candlestick patterns often show limited predictive power due

to low volatility and strong regulatory oversight. Studies in Mexico and China have Highlighted the challenges of applying technical analysis in these sectors. The technology sector in emerging markets is characterized by High growth and volatility, making it ideal for candlestick-based strategies. Research in China and India has demonstrated the effectiveness of patterns like Hammer and Shooting Star in this sector (Bulkowski, 2021).

2.4 Challenges in Applying Technical Analysis in Low-Liquidity Markets

Candlesticks, like every other tool, come with their own set of challenges. These challenges can be intrinsic to the tool itself, or limitations imposed by the users. Candlestick patterns have shown varying degrees of success in emerging markets, depending on factors such as sector characteristics, market liquidity, and external influences. While these patterns are effective in High-volatility, High-liquidity sectors like banking and technology, their predictive power diminishes in stable, low-liquidity sectors. Additionally, challenges such as false signals, market manipulation, and data quality issues Highlight the need for caution when applying technical analysis in emerging markets. Here are the major challenges that affect the use of candlestick patterns:

False Signals

In low-liquidity markets, price movements are often driven by a small number of trades, leading to false signals and exaggerated patterns. For example, a Hammer pattern in a low-liquidity stock may not indicate a genuine reversal but rather a temporary imbalance in supply and demand.

Market Manipulation

Emerging markets are often susceptible to market manipulation, where large traders or institutions artificially inflate or deflate prices. This can distort candlestick patterns and reduce their reliability. For instance, studies in Nigeria have shown that patterns like Bullish Engulfing are often exploited by manipulators to lure retail investors into unfavorable trades (Olowe, 2009).

External Factors

In many emerging markets, external factors such as political instability, currency fluctuations, and commodity prices play a significant role in driving price movements.

These factors can override the signals provided by candlestick patterns, making them less effective. For example, in South Africa, the performance of candlestick patterns in the mining sector is heavily influenced by global commodity prices (Kalidas et al., 2013).

Data Quality

The quality and availability of historical price data in emerging markets can be inconsistent, making it difficult to conduct reliable technical analysis. For instance, in smaller African markets, data gaps and inconsistencies often limit the applicability of candlestick patterns (Olowe, 2009).

2.5 Nepali Stock Market Dynamics

This section provides an in-depth analysis of the NEPSE, focusing on its historical performance, investor behavior, and existing research on technical analysis. Below is a detailed expansion of each subsection:

2.5.1 Historical Performance of NEPSE

The NEPSE was established in 1993 and is the sole stock exchange in Nepal. It serves as the primary platform for trading equities, bonds, and other securities. As of 2023, NEPSE has a market capitalization of approximately USD 20 billion, with over 200 listed companies spanning sectors such as banking, hydropower, microfinance, and manufacturing (Nepal Rastra Bank, 2023). NEPSE has experienced significant volatility over the years, driven by factors such as political instability, regulatory changes, and macroeconomic shocks. For example, the market saw a sharp decline during the 2015 earthquake and the COVID-19 pandemic but rebounded strongly in 2021 due to increased retail participation and favorable government policies.

Key milestones in NEPSE's history include:

2008: Introduction of automated trading, which improved market efficiency and transparency.

2016: Implementation of the NEPSE Index, tracking performance of all listed companies.

2021: Record-High NEPSE Index levels, driven by a surge in retail investor activity.

NEPSE is characterized by High volatility, with frequent price swings driven by speculative trading and external factors. For instance, the NEPSE Index experienced a 52-

week High of 3,200 points and a 52-week low of 1,800 points in 2022, reflecting the market's sensitivity to news and investor sentiment.

2.5.2 Key Sectors

Banking: The banking sector dominates NEPSE, accounting for nearly 60% of total market capitalization. It is Highly liquid and attracts significant institutional participation.

Hydropower: A growth sector driven by government policies and infrastructure development. Hydropower stocks are often volatile, with prices influenced by regulatory changes and project approvals.

Microfinance: Sensitive to interest rate fluctuations and regulatory shifts. Microfinance stocks are popular among retail investors due to their High dividend yields.

Manufacturing and Trading: These sectors have lower liquidity and are often overlooked by institutional investors, making them prone to speculative trading.

2.5.3 Behavioral Finance in Nepal

Retail investors account for over 90% of trading activity in NEPSE, making it one of the most retail-driven markets in the world (Poudel et al., 2024) . This dominance is attributed to the lack of institutional participation and limited access to alternative investment options. Retail investors in Nepal often rely on informal information sources, such as rumors, social media, and peer recommendations, rather than fundamental or technical analysis. Herd behavior is prevalent in NEPSE, with investors often following the actions of others rather than making independent decisions. This behavior is driven by factors such as:

Lack of financial literacy: Many retail investors lack the knowledge and skills to analyze market trends.

Fear of missing out (FOMO): Investors rush to buy stocks that are trending upward, often leading to bubbles.

Overreaction to news: Positive or negative news can trigger exaggerated price movements, as investors react impulsively.

Overconfidence: Investors overestimate their ability to predict market movements, leading to excessive trading.

Anchoring: Investors rely too heavily on historical prices when making decisions, ignoring new information.

Confirmation bias: Investors seek information that confirms their existing beliefs, leading to poor decision-making.

2.5.4 Existing Research on Technical Analysis in Nepal

Behavioral Finance and Market Efficiency in Nepal, explored the impact of investor behavior on market efficiency. The study found that retail investors' reliance on technical analysis tools like candlestick patterns was limited, with most investors preferring fundamental analysis or informal methods. However, the study also noted that technical analysis could be effective in NEPSE, given the market's High volatility and retail-driven nature. It also Highlighted the prevalence of herd behavior and overreaction to news among Nepali investors (Poudel et al., 2024). Despite the growing popularity of technical analysis in global markets, there is a lack of empirical research on its efficacy in Nepal. Existing studies have focused primarily on investor behavior and market efficiency, with little attention paid to specific technical tools like candlestick patterns. This gap underscores the need for research that evaluates the performance of candlestick patterns in NEPSE, particularly in High-liquidity sectors like banking and hydropower. The study suggested that technical analysis tools like candlestick patterns could help mitigate these biases by providing a systematic framework for decision-making.

NEPSE's unique characteristics High retail participation, low institutional involvement, and sectoral diversity make it an ideal setting for studying the efficacy of technical analysis tools like candlestick patterns. By addressing these challenges and leveraging the market's unique dynamics, this study aims to provide valuable insights for retail investors, academics, and policymakers in Nepal.

2.6 Knowledge Gap and Justification for the Study

Despite the growing popularity of technical analysis tools like candlestick patterns in global markets, there is a notable lack of empirical research on their efficacy in Nepal. Existing studies, such as Sharma's Behavioral Finance and Market Efficiency in Nepal, have focused primarily on investor behavior and market efficiency, with little attention paid

to specific technical tools. This gap is particularly concerning given the unique characteristics of Nepal's stock market, including its High retail participation, low institutional involvement, and sectoral diversity (Poudel et al., 2024).

Most research on candlestick patterns has been conducted in developed markets (e.g., the U.S., Europe) and larger emerging markets (e.g., India, China). While these studies provide valuable insights, their findings may not be directly applicable to smaller, less efficient markets like Nepal. For example, studies in India have shown that candlestick patterns are effective in High-volatility sectors like technology, but similar research is lacking in Nepal's context (Sehgal & Gupta, 2007). Existing research on technical analysis in Nepal has largely ignored sector-specific dynamics and the impact of holding periods on the performance of candlestick patterns. This oversight is problematic because the efficacy of these patterns may vary across sectors and timeframes due to differences in liquidity, volatility, and investor behavior.

This study addresses the gap by analyzing the performance of candlestick patterns across key sectors in Nepal, including banking, hydropower, microfinance, and low-float stocks. By focusing on sector-specific dynamics, the study provides a nuanced understanding of how these patterns perform in different market segments. For example, the study will examine whether candlestick patterns are more effective in High-liquidity sectors like banking, where institutional participation is Higher, or in low-liquidity sectors like hydropower, where retail speculation dominates. This study also addresses the temporal gap by evaluating the performance of candlestick patterns over three distinct holding periods: 2, 4, and 8 days. This analysis will reveal whether shorter or longer holding periods yield better results and provide insights into the time sensitivity of candlestick patterns. For instance, the study will test the hypothesis that shorter holding periods (e.g., 2 days) show greater predictive accuracy, as candlestick patterns are most effective in capturing short-term price reversals.

Chi-square testing provides an objective measure of performance, reducing the risk of bias or subjective interpretation. In this study, chi-square testing will be used to compare the

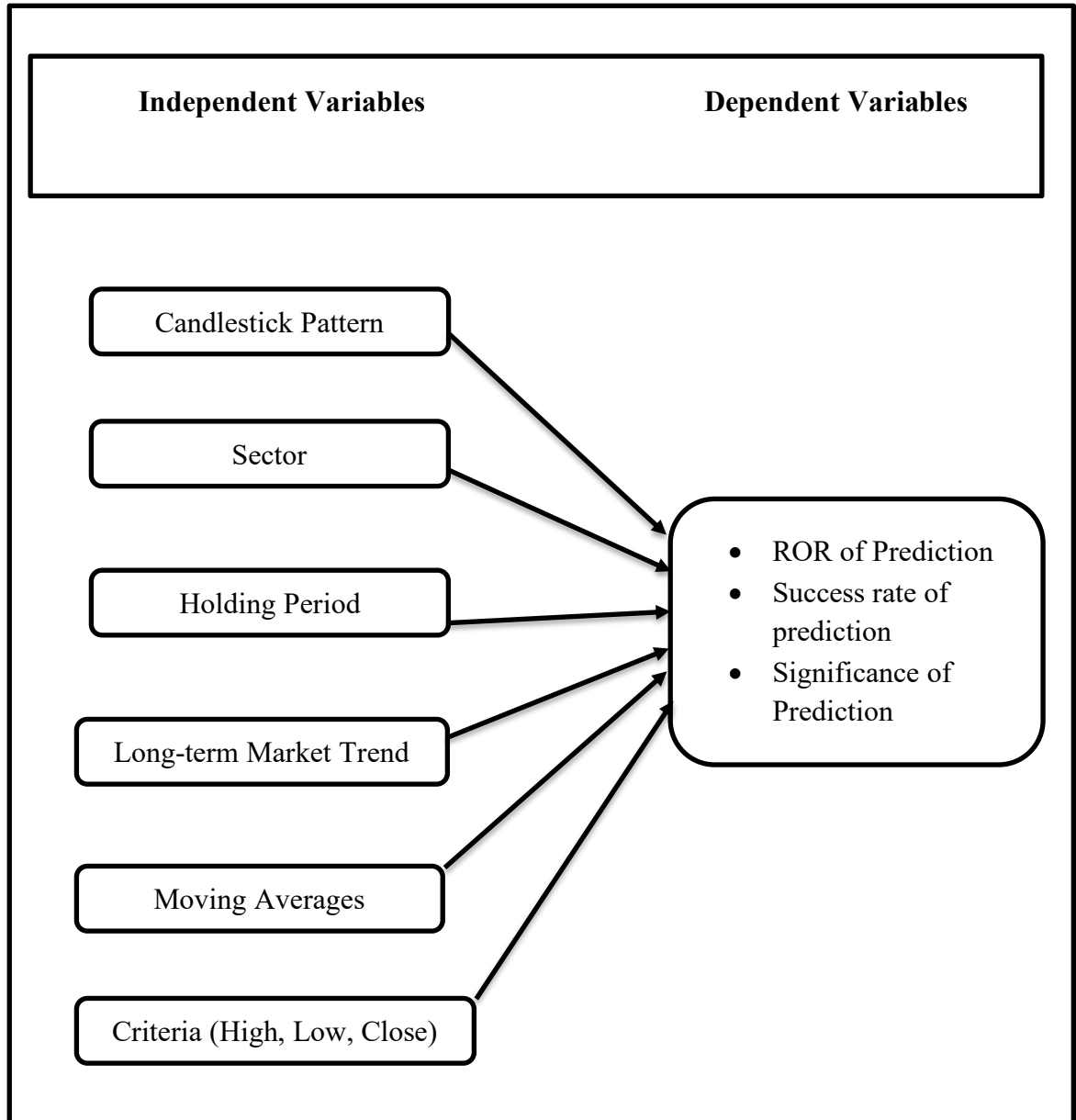
performance of candlestick patterns (e.g., Hammer, Shooting Star) to random stock picks across different sectors and holding periods. The test attempts to evaluate the predictive power of these patterns and identifying any statistically significant differences. For instance, the study will use chi-square tests to determine whether the success rate of the Hammer pattern in the banking sector is significantly Higher than that of random picks over a 2-day holding period. For example, if candlestick patterns have a success rate of 65% in predicting price reversals, while random picks have a success rate of 50%, the Chi-square Test can determine whether this difference is due to chance or reflects a genuine advantage of candlestick patterns.

By comparing the performance of candlestick patterns to random stock picks, the study attempts to provide a robust benchmark for evaluating their predictive power. This comparative approach ensures that the findings are grounded in empirical evidence and not influenced by bias, subjective interpretation and market-wide trends or external factors.

2.7 Conceptual Framework

This study is premised on the hypothesis that specific characteristics inherent to candlestick charts and the surrounding market conditions can significantly influence the predictive accuracy regarding future stock movements. To systematically investigate this hypothesis, the framework identifies several independent variables and outlines how they are expected to affect the key outcome measures namely, the rate of return, the success rate, and the statistical significance of candlestick prediction success.

Figure 10: *Independent and Dependent Variables*



CHAPTER III

RESEARCH METHODOLOGY

3.1 Research Design

This Section aims to Highlight the design and methodology used to obtain required data. It explains thoroughly about the research design, data collection methods, sampling design, research instrument, constructs measurement, data processing and methods of data analysis.

A research design is a structured plan guiding how to collect, measure, and analyze data to effectively answer a research question. It acts as a blueprint to ensure accuracy, reliability, and efficiency while minimizing bias, errors, and time wastage. Key benefits include organizing research materials, testing hypotheses, estimating required resources, referencing expert insights, and directing the study toward clear, valid outcomes. Essentially, it streamlines the research process to achieve dependable and focused results. (Ahmad, 2021).

This study adopts a comparative quasi-experimental design to evaluate the prediction accuracy of candlestick patterns against random stock selection in Nepal's stock market. The reason for selection this research design is because we cannot by ourselves randomize the data of stock market as we are doing our experiment in historical data. On top of that randomizing any stock's trading points isn't possible and even if it was its not ethical and legal. For that reason, we are taking randoms from existing real data. By analyzing historical price data (2019–2024), the research compares two strategies:

1. **Candlestick Strategy:** Selection of stocks based on the occurrence of Hammer, Shooting Star, Inverted Hammer, and Hanging Man patterns.
2. **Random Strategy:** Selection of stocks on arbitrary dates, matched by sector and year to control for external market trend.

The study will attempt to explain the result using descriptive analysis of the success rate of the candle stick's prediction and infernal data analysis using p-value to establish the significance of the findings from the descriptive analysis.

3.2 Population, Sample and Sampling Strategy

For this study, the target population is all the stocks listed in NEPSE. But to make the calculation and analysis process significant yet feasible, a sample of stocks is picked. 5 Category of stocks are studied with 15 stocks being covered by each category. Follows are the sectors of focus on this study:

Banking: Dominates NEPSE (60% market cap) with High liquidity (NEPSE Annual Report, 2023).

Hydropower: Emerging sector with High volatility due to regulatory shifts.

Microfinance: Low-floating shares and retail investor dominance.

High Price Stocks: Defined as stocks trading above NPR 2,100 per share.

Low Floating/Low Cap Share Stocks: Less than 20% or 50 crores worth of shares are available for public trading.

The sample dataset spans from mid-2019 to mid-2024 to capture various market conditions, including bullish, bearish, and sideways trends. Data are collected covering NEPSE price history of 15 randomly picked stocks for 5 categories each. The full list of stocks picked and their category is presented in following table:

Table 1: *Stocks under study*

Banks	Hydropower	Microfinance	Expensive	Low Cap Stocks
NABIL	NHPC	NMBMF	UNL	SAMAJ
NICA	UPPER	CBBL	RBCL	SMH
SCB	AHPC	DDBL	BNT	SPL
HBL	API	SKBBL	STC	KBSH
EBL	SHPC	SWBBL	BBC	BHPL
GBIME	NGPL	GRDBL	NLIC	ANLB
MBL	HPPL	NUBL	NLICL	WNLB
SBL	SIKLES	MLBS	SHIVM	DLBS
PRVU	RADHI	FOWAD	NRIC	SMB
CZBIL	AKJCL	SADBL	SHL	MLBS
LBL	MKHC	MERO	NTC	SHLB
ADBL	BARUN	KMCDB	SICL	SMFBS
SBI	TVCL	SLBBL	BNL	UNLB
SANIMA	USHL	MLBSL	ANLB	GMFBS
KBL	KBSH	JALPA	MLBSL	JSLBB

3.3 Data Collection Method and Tools

This research is based on secondary data sources. The required data was sourced from NepseAlpha (<https://www.nepsealpha.com>), which provides comprehensive historical stock price data for companies listed on the NEPSE. This dataset includes daily trading information with opening, closing, High, low, and trading volumes. Data are downloaded one by one using the manual download panel NEPSE Price Export web portal. Once the data are downloaded in JSON format, they are ready to be cleaned up, transformed and analysed.

3.4 Data Analysis Tools and Techniques

Various statistical tools can be used to analyze the available financial data. These tools are used in research to draw reliable conclusion through the analysis of financial data. candlestick patterns detection with custom python code, scipy, pandas and plotlyx are used to compute and visualize data using python programming language.

3.4.1 Algorithms used for candlestick patterns

Candlestick patterns are constructed using HLOC data of one or more days in a time series data of the stock's price history. Body position (upper/lower third of TotalRange) determines trend context. Trend confirmation (prior uptrend/downtrend) is critical for pattern validity. Followings are variables we use to detect candlestick patterns

O = Open price

H = High price

L = Low price

C = Close price

Body(B) = $|C - O|$ (*absolute value of close minus open*)

UpperShadow(US) = $H - \max(O, C)$ (*distance from High to the top of the body*)

LowerShadow(LS) = $\min(O, C) - L$ (*distance from low to the bottom of the body*)

TotalRange(R) = $H - L$ (*total price range of the candle*)

For our study, we are only considering one day patterns. Algorithm for detecting each patterns is described below:

Hammer: A bullish reversal signal after a downtrend. Small body at the upper end of the candle, long lower shadow ($\geq 2 \times$ body), minimal/no upper shadow. It's detection criteria is:

$$LS \geq 2 * B \ \&\& \ US \leq B \ \&\& \ B \leq 0.2 * R$$

Inverted Hammer: A bullish reversal signal after a downtrend. Small body at the lower end, long upper shadow ($\geq 2 \times$ body), minimal lower shadow.

$$US \geq 2 * B \ \&\& \ LS \leq B \ \&\& \ B \leq 0.2 * R$$

Shooting Star: A bearish reversal signal after an uptrend. Small body at the lower end, long upper shadow ($\geq 2 \times$ body), minimal lower shadow.

$$US \geq 2 * B \ \&\& \ LS \leq B \ \&\& \ B \leq 0.2 * R$$

Hanging Man: a bearish reversal signal after an uptrend. Small body at the upper end, long lower shadow ($\geq 2 \times$ body), minimal/no upper shadow.

$$LS \geq 2 * B \ \&\& \ US \leq B \ \&\& \ B \leq 0.2 * TR$$

3.4.2 Return Rate

For each pattern, return rate on Nth day after holding for n days is calculated by comparing the candlestick day's criteria (c) with moving average of n days criteria (c), Return Rate R_{NC} is calculated as:

$$R_{c(N+n)} = \frac{MA_{c(N+n)} - c_N}{c_N} \times 100\%$$

3.4.3 Success Rate

Success rate or hit ratio is calculated by comparing total number of candles that successfully predict the incoming trend $N(success)$ divided by total number of candles under observation $N(total)$. Success rate for n day holding period on criteria c is for that candle is given by:

$$Hit_{nc} \% = \frac{N(success)_{nc}}{N(Total)_{nc}}$$

3.4.4 Moving Average

The n day moving average of a stock on a criteria c is given by:

$$MA_{nc} = \frac{\sum_{i=N}^{N+n} (C_{N+i})}{n}$$

3.4.5 Chi-square Test and P-Value

Instances, where candlestick performance underperformed with respect to random benchmarks, were excluded to mitigate noise and focus on statistically significant cases. Using SciPy in python, we can compute Chi-square Test with p value and degree of freedom using:

Figure 11: *Python Code to Compute Chi-square and P - Value*

```
import scipy.stats as stats
import numpy as np

# Calculate contingency table using variables of candle and
candlesticks

contingency_table = np.array([[x1, x2], [y1, y2]])

chi_square, p_value, d_o_freedom = stats.chi2_contingency(observed)
```

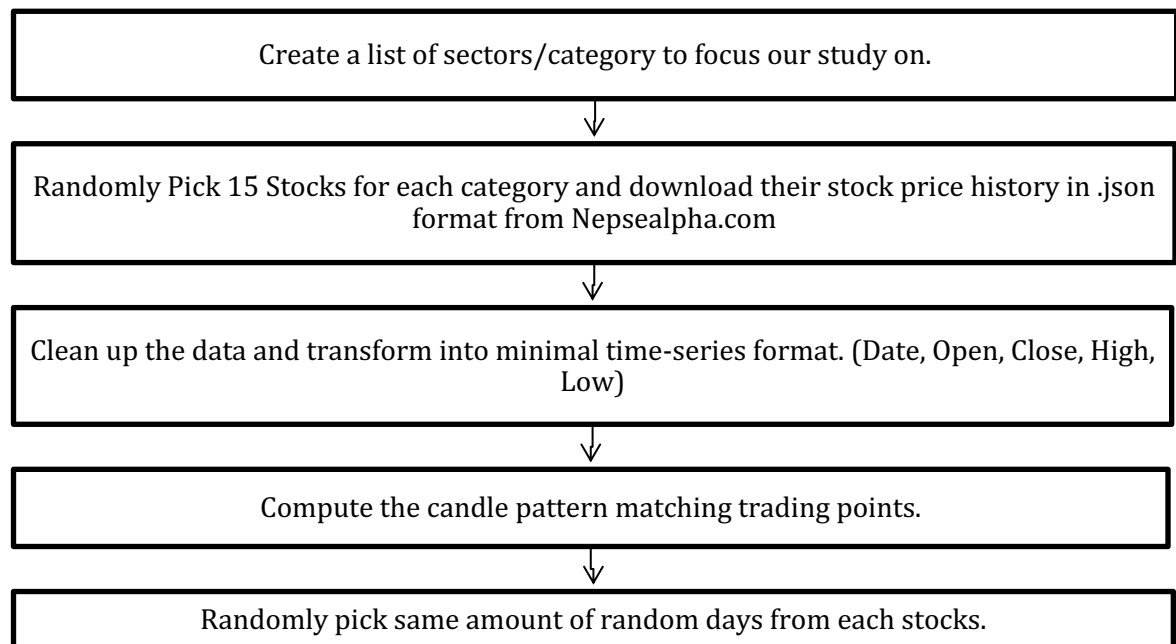
3.5 Data Collection procedure

In the initial stage of the study, I was attempting to compare the performance of candle stick charts in US market vs Nepali Market. I downloaded data of top 200 companies from yahoo finance using yfinance python library and manually downloaded all the stock's price history time series data from nepsealpha.com. I started to detect patterns in the time series using Ta-Lib and pandas-ta libraries in python, and went on to validate the occurrence of the candlestick on real data on tradingview.com. It didn't give me satisfactory result, mainly because these libraries made few assumptions during the detection process, which I didn't want to include in my research as they were excluding candles outside of the favourable shortterm trend, made assumptions about the size of body and whiskers in order to qualify into a candle pattern which was causing both false positives and absence of candlestick patterns I wanted to include in my study.

Next, I implemented my own version of candlestick pattern recognition algorithm to detect these patterns in a format and assumptions I was trying to test my hypothesis against. Then I had to drop the idea of comparing candlesticks performance of NEPSE against US markets i.e. S&P500 because it would just compare the performance of candlestick

performance of US against Nepali market, i.e. comparison of two markets on how they favor candlesticks instead of the efficacy of the candlesticks themselves which may not be of much use. So, I pivoted my study towards comparing the efficacy of using candlesticks for decision making inside the Nepali Market. For that, I randomly selected the same amount of price points from each stocks that there were for detected candles, made sure they were a valid timeseries data, followed same data structure. Similar technique of comparing candlestick's performance with random points was applied in another study of candlestick's performance in S&P500 (Jamaloodeen et al., 2018). Now the data was ready for our next stage i.e. descriptive data analysis. The process of data collection is presented in the following figure in brief.

Figure 12: *Data Collection Process*



3.6 Ethical Considerations

My study is based on secondary data available on public domain on the internet for use and download. For that reason, I can confirm the data for the study being sourced while being legally, ethically harmless to any parties. On top of that as I am not using it for any business

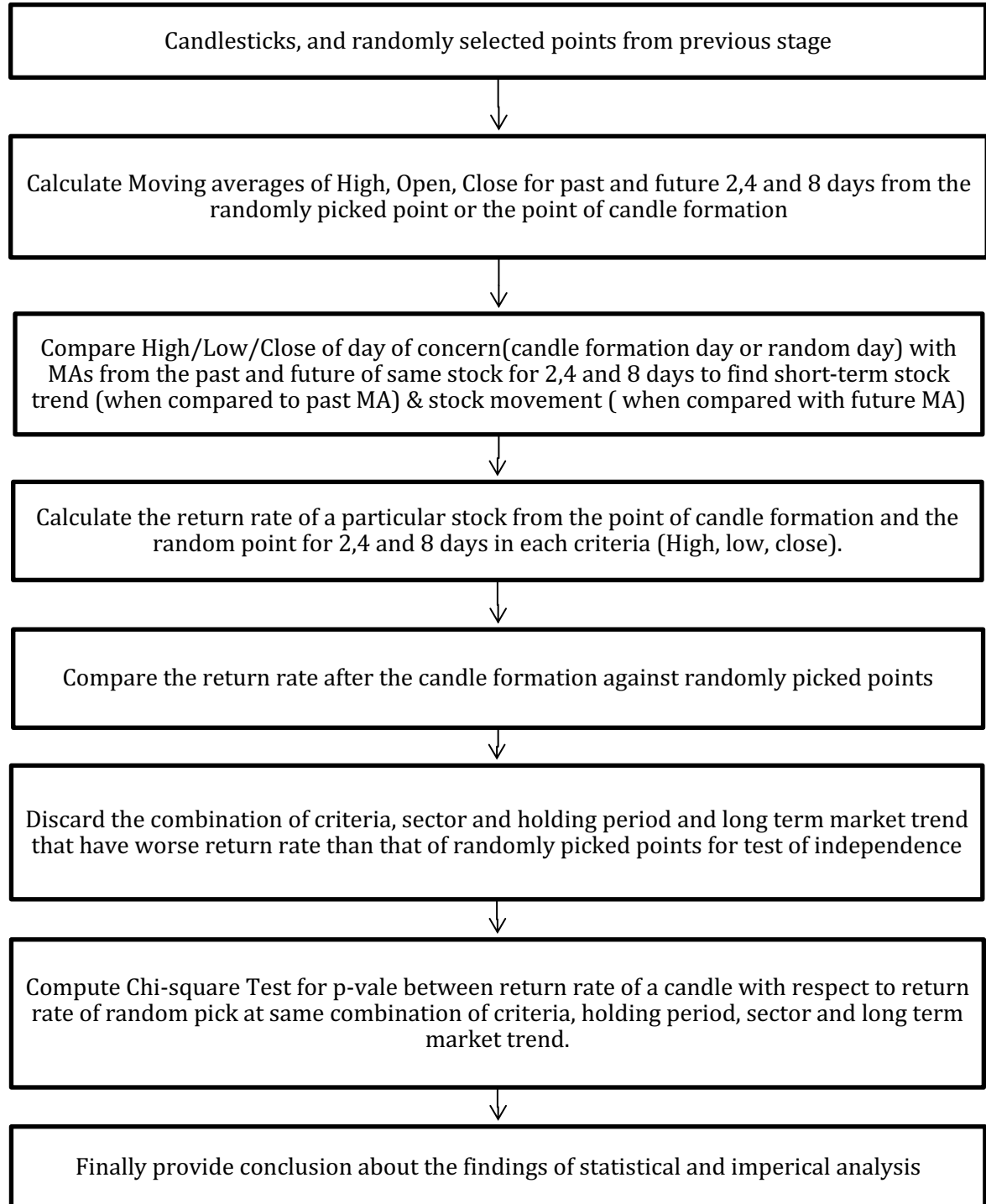
or monetary purpose, it doesn't require any explicit authorization from data provider for the data in public domain.

3.7 Data Analysis and Interpretation

This is the stage where the statistical and inferential analysis was done on the data. First, we categorized the data based on long term trend (Bullish/Bearish) and sectors. We created a study scope on 10 different combinations where there were two long term trends and 5 sectors of study. Secondly, Based on the data analysis tools described on section 3.4, we generated various statistical parameters like Moving Average, Return Rate, Success Rate for each category for 2,4 and 8 days holding period in long term bullish and bearish markets in 5 different sectors. In each scope we studied the success rates of prediction after 2,4 and 8 days for 4 different candles. Then we compared the success rate with the success rates of randomly selected points against candlestick points. This gave us a rough idea about which candle gives respectable results on which scope.

But to further prove the validity of these observation of each scopes as being a significantly better than randomly selected points, Chi-square Test of independence was needed between success rate of random stock selection and candlesticks on each scopes. For that we couldn't include the combinations where success rate of candles that was worse than random picks. The reason behind the need for discarding these values is that if a candlestick is significantly worse at predicting a stocks movement direction, it would float on the surface as significant difference with respect to random selection. On top of that, we could discard the scopes for each candles that even descriptive analysis declares as being nowhere close to produce better result than random selection and focus on scopes. We compared the success rate of the candle for each combination. Finally, we would perform Chi-square Test for p-value of two distributions and get the result of whether a candle's prediction on each scope were significantly better than random prediction. Note: on chi-square test, the test will only tell whether candle's prediction is significantly different from random selection. But as only the combination of candlesticks and scope that yielded better returns than random selection reached the point of chi-square, the result of chi-square test's p-value gives on the confidence about the prediction being significantly better. The overall analysis process is presented in the figure below.

Figure 13: *Data analysis flow chart*



CHAPTER IV

DATA PRESENTATION AND ANALYSIS

4.1 Descriptive Analysis

This section is dedicated to exploring and compare the success rate of each candlestick. Each table will contain the comparison of candlesticks under 2, 4 and, 8 days holding period, whether they were in expected short term trend or not and over 3 criteria namely Close, High and Low.

Table 2: *Candles Observation for Banking Stocks in Bullish Market*

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	57	114	79.0	81.6	71.9	72.8	47.4	50.9
	I. Hammer	107	232	42.1	39.2	40.2	38.4	57.0	53.5
	Random	614	1454	49.2	48.0	47.9	47.9	48.5	47.3
	Random*	660	1454	45.5	49.5	46.1	48.6	46.7	49.4
	Hanging Man	101	233	25.7	25.8	31.7	31.3	42.6	45.5
	Shooting Star	62	136	69.3	69.8	62.9	65.4	38.7	41.2
4D	Hammer	59	114	67.8	74.6	69.5	70.2	45.8	46.5
	I. Hammer	109	232	53.2	51.3	53.2	51.3	62.4	59.9
	Random	624	1454	48.7	48.4	50.6	50.3	50.3	50.9
	Random*	616	1454	50.3	50.5	48.7	48.4	49.0	48.3
	Hanging Man	97	233	30.9	33.5	30.9	37.3	48.5	48.9
	Shooting Star	52	136	53.9	55.9	53.9	59.6	42.3	41.2
8D	Hammer	44	114	68.2	71.0	68.2	69.3	52.3	49.1
	I. Hammer	98	232	52.0	51.3	57.1	54.3	62.2	58.2
	Random	620	1454	45.2	49.1	46.5	51.6	46.1	51.0
	Random*	520	1454	60.4	49.0	59.2	46.4	60.0	47.3
	Hanging Man	83	233	33.7	34.3	42.2	39.1	49.4	48.1
	Shooting Star	43	136	55.8	47.8	58.1	52.2	48.8	37.5

Source: Author's Calculation

Table 2 examines the predictive capabilities of candlestick patterns in the Banking sector during a sustained Bullish market phase. Hammer displays significantly better results at forecasting High criteria for two days and four days. It also renders slightly better results at forecasting low criteria for two days. Additionally, it shows slightly better at forecasting High criteria (following a trend) for four days and High criteria and low criteria (following a trend) for eight days.

Inverted Hammer yields slightly better results at forecasting closing prices (following a trend) for eight days. But, Hanging Man shows worse results at forecasting High criteria, low criteria for two days, and High criteria and low criteria (following a trend) for four days and High criteria and low criteria (following a trend) for eight days when compared to a random selection of stocks. Shooting Star yields slightly better results at forecasting High criteria and low criteria for two days.

Table 3: *Candles Observation for Banking Stocks in Bearish Market*

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	46	111	87.0	81.1	80.4	73.9	39.1	43.2
	I. Hammer	132	220	30.3	26.4	30.3	26.8	43.2	37.3
	Random	528	1412	34.1	39.5	38.6	39.0	37.1	39.4
	Random*	758	1412	44.9	59.4	40.6	59.4	43.5	59.9
	Hanging Man	92	247	38.0	34.8	35.9	36.0	57.6	61.1
	Shooting Star	37	140	81.1	76.4	78.4	78.6	56.8	47.9
4D	Hammer	57	111	63.2	68.5	63.2	64.0	38.6	40.5
	I. Hammer	144	220	28.5	27.7	29.2	28.6	42.4	40.0
	Random	418	1412	30.1	38.2	34.5	38.2	31.6	39.4
	Random*	780	1412	37.4	61.5	34.6	61.2	36.7	60.3
	Hanging Man	76	247	36.8	47.0	36.8	45.8	57.9	63.2
	Shooting Star	40	140	70.0	69.3	72.5	71.4	60.0	52.9
8D	Hammer	65	111	53.9	56.8	49.2	52.2	29.2	31.5
	I. Hammer	119	220	31.1	30.9	26.1	27.7	35.3	35.9
	Random	324	1412	38.3	37.0	41.4	36.0	41.4	39.4
	Random*	826	1412	24.2	62.8	23.0	63.7	23.0	60.1
	Hanging Man	44	247	47.7	55.1	47.7	55.1	56.8	64.8
	Shooting Star	37	140	64.9	65.0	62.2	72.1	43.2	52.9

Source: Author's Calculation

Table 3 analyzes the predictive power of candlestick patterns in the Banking sector during a prolonged Bearish market phase. The Hammer pattern consistently performs better in forecasting Highs, Lows for two days, and Highs and Lows for four days. It also shows slightly improved results in forecasting Highs and Lows for eight days. In contrast, the Inverted Hammer pattern yields worse results in forecasting Lows (in accordance with a trend) for eight days compared to random selection of stocks.

The Hanging Man pattern consistently performs better in forecasting Closes (in accordance with a trend) for eight days. It also shows slightly improved results in forecasting Closes

(in accordance with a trend) for four days and Highs (in accordance with a trend) and Lows (in accordance with a trend) for eight days. However, it yields worse results in forecasting Highs, Lows for two days, and Lows for four days compared to random selection of stocks. The Shooting Star pattern consistently performs better in forecasting Highs (in accordance with a trend), Lows (in accordance with a trend) for two days, Highs (in accordance with a trend), Lows (in accordance with a trend) for four days, and Highs (in accordance with a trend), Lows (in accordance with a trend) for eight days. It also yields slightly improved results in forecasting Highs and Lows for two days and Closes (in accordance with a trend) for four days and Closes (in accordance with a trend) for eight days.

Similarly, Table 4 analyzes the predictive power of candlestick patterns in the Hydropower sector during a prolonged Bullish market phase. The Hammer pattern demonstrates superior accuracy in forecasting High prices (following a trend) for two days and High prices (following a trend) and low prices (following a trend) for eight days. It also shows slightly better results in forecasting High prices and low prices (following a trend) for two days and High prices and low prices (following a trend) for four days.

The Inverted Hammer pattern yields slightly better results in forecasting closing prices (following a trend) for four days and closing prices (following a trend) for eight days.

The Hanging Man pattern shows slightly better results in forecasting closing prices (following a trend) for two days. However, it performs worse in forecasting High prices (following a trend) for four days, High prices (following a trend), low prices (following a trend), and closing prices (following a trend) for eight days when compared to a random selection of stocks. The Shooting Star pattern demonstrates significantly better accuracy in forecasting High prices (following a trend) and low prices for two days.

Table 4: *Candles Observation for Hydropower Stocks in Bullish Market*

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	38	93	84.2	73.1	76.3	72.0	47.4	41.9
	I. Hammer	127	243	43.3	44.4	44.9	44.0	59.1	59.7
	Random	534	1182	44.9	49.4	52.1	53.5	48.7	49.9
	Random*	572	1182	49.0	47.4	40.6	42.0	43.7	45.7
	Hanging Man	77	147	42.9	38.8	42.9	33.3	62.3	55.1
	Shooting Star	53	134	75.5	68.7	83.0	76.1	49.1	41.8
4D	Hammer	37	93	73.0	68.8	70.3	66.7	46.0	43.0
	I. Hammer	107	243	43.9	49.8	49.5	49.8	65.4	63.0
	Random	600	1182	48.3	50.2	52.7	52.8	49.0	50.8
	Random*	458	1182	63.8	47.4	57.2	44.2	62.5	46.4
	Hanging Man	73	147	39.7	40.8	42.5	38.8	52.0	51.7
	Shooting Star	59	134	66.1	61.2	74.6	67.2	47.5	41.0
8D	Hammer	32	93	84.4	62.4	81.2	63.4	40.6	40.9
	I. Hammer	102	243	57.8	55.1	60.8	54.3	70.6	64.6
	Random	656	1182	50.3	51.6	53.4	54.5	53.0	54.0
	Random*	348	1182	80.5	44.3	74.7	40.8	75.9	41.8
	Hanging Man	75	147	41.3	42.9	38.7	36.7	52.0	53.7
	Shooting Star	69	134	59.4	55.2	60.9	61.2	44.9	33.6

Source: Author's Calculation

It also shows slightly better results in forecasting High prices for two days and low prices for four days and low prices for eight days. Additionally, it yields worse results in forecasting High prices (following a trend), closing prices (following a trend), and closing prices (following a trend) for eight days when compared to a random selection of stocks.

Table 5. Candles Observation for Hydropower Stocks in Bearish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	39	74	64.1	58.1	59.0	55.4	41.0	35.1
	I. Hammer	106	171	34.9	33.3	34.0	33.3	46.2	45.6
	Random	420	1036	33.3	37.5	39.0	41.3	44.8	45.4
	Random*	580	1036	47.6	61.2	43.1	57.7	39.7	54.2
	Hanging Man	50	114	50.0	36.8	56.0	43.0	72.0	56.1
	Shooting Star	39	133	76.9	74.4	74.4	79.0	59.0	58.6
4D	Hammer	41	74	53.7	52.7	51.2	55.4	36.6	36.5
	I. Hammer	97	171	33.0	35.7	33.0	33.9	41.2	44.4
	Random	362	1036	35.4	37.1	46.4	41.3	44.2	43.8
	Random*	588	1036	39.8	62.7	33.0	58.7	34.4	56.2
	Hanging Man	51	114	49.0	45.6	47.1	52.6	58.8	55.3
	Shooting Star	40	133	80.0	72.9	82.5	79.0	62.5	58.6
8D	Hammer	48	74	50.0	54.0	45.8	51.4	35.4	40.5
	I. Hammer	98	171	40.8	35.7	34.7	31.6	45.9	41.5
	Random	346	1036	33.0	38.2	41.0	39.0	39.3	40.0
	Random*	590	1036	39.3	61.8	34.6	61.0	35.6	60.0
	Hanging Man	36	114	50.0	51.8	50.0	50.0	58.3	56.1
	Shooting Star	44	133	79.5	72.2	77.3	74.4	77.3	62.4

Source: Author's Calculation

Table 5 analyzes the predictive power of candlestick patterns in the Hydropower sector during a prolonged Bearish market phase. The Hammer pattern consistently outperforms other patterns in forecasting Highs (following a trend) for two days. It also shows slightly improved results in forecasting Highs and Lows (following a trend) for two days. Additionally, it demonstrates slightly better accuracy in forecasting Highs for four days and Highs for eight days.

The Hanging Man pattern exhibits significantly better results in forecasting Closes (following a trend) for two days. It also shows slightly improved accuracy in forecasting

Closes and Lows (following a trend) for four days. Furthermore, it shows slightly better results in forecasting Closes for eight days. However, it performs worse in forecasting Highs for two days and four days compared to a random selection of stocks.

The Shooting Star pattern consistently outperforms other patterns in forecasting Highs (following a trend) and Lows (following a trend) for two days. It also demonstrates slightly improved accuracy in forecasting Highs, Lows, and Closes (following a trend) for four days. Additionally, it shows slightly better accuracy in forecasting Highs, Lows, and Closes (following a trend) for eight days.

Table 6. Candles Observation for Micro Finance Stocks in Bullish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	52	134	75.0	77.6	71.2	75.4	46.1	49.2
	I. Hammer	150	285	44.0	41.0	46.0	45.3	56.0	51.2
	Random	866	1718	49.0	51.3	54.7	54.5	52.7	52.6
	Random*	704	1718	60.2	46.1	52.6	43.2	56.2	45.3
	Hanging Man	118	238	28.0	34.0	24.6	31.5	44.9	52.1
	Shooting Star	92	198	55.4	60.6	69.6	69.7	42.4	45.5
4D	Hammer	48	134	72.9	73.9	75.0	75.4	54.2	57.5
	I. Hammer	147	285	47.6	46.7	49.0	48.8	55.8	56.5
	Random	894	1718	51.9	53.7	53.9	54.7	51.7	53.9
	Random*	640	1718	65.9	44.5	63.4	43.2	65.6	44.1
	Hanging Man	105	238	33.3	33.2	35.2	36.5	45.7	47.5
	Shooting Star	101	198	54.5	55.6	53.5	57.1	41.6	41.9
8D	Hammer	57	134	71.9	76.9	75.4	80.6	56.1	58.2
	I. Hammer	119	285	59.7	53.7	58.0	53.7	63.0	58.6
	Random	884	1718	53.2	54.0	54.5	56.2	54.5	54.8
	Random*	548	1718	72.6	43.7	70.1	41.6	70.4	43.1
	Hanging Man	102	238	31.4	30.2	35.3	33.6	38.2	39.9
	Shooting Star	99	198	51.5	50.5	43.4	48.0	35.4	37.4

Source: Author's Calculation

Table 6 analyzes the predictive power of candlestick patterns in the Microfinance sector during a prolonged bullish market phase.

The Hammer pattern consistently outperforms other patterns in forecasting High prices for two consecutive days. It also shows slightly better results in forecasting low prices for two consecutive days. Additionally, it demonstrates slightly better accuracy in forecasting High and low prices for four consecutive days and High and low prices for eight consecutive days compared to random selection of stocks.

The Hanging Man pattern, on the other hand, performs poorly in forecasting High prices (following a trend) and low prices (following a trend) for two consecutive days. Furthermore, it shows slightly worse accuracy in forecasting High prices (following a trend), low prices (following a trend), and closing prices (following a trend) for four consecutive days and High prices (following a trend), low prices (following a trend), and closing prices (following a trend) for eight consecutive days compared to random selection of stocks.

The Shooting Star pattern consistently outperforms other patterns in forecasting low prices for two consecutive days. It also shows slightly better accuracy in forecasting low prices (following a trend) for two consecutive days. However, it demonstrates slightly worse accuracy in forecasting closing prices (following a trend) for four consecutive days and High prices (following a trend), low prices (following a trend), and closing prices (following a trend) for eight consecutive days compared to random selection of stocks.

Table 7 analyzes the predictive power of candlestick patterns in the Microfinance sector during a prolonged Bearish market phase.

The Hammer pattern performs exceptionally well in forecasting Highs (in accordance with a trend) for two consecutive days. It also yields slightly better results in forecasting Highs, Lows, and Highs for four consecutive days. However, it performs poorly in forecasting Closes (in accordance with a trend) for two consecutive days when compared to random selection of stocks.

The Hanging Man pattern shows promising results in forecasting Closes (in accordance with a trend) for four consecutive days and for eight consecutive days. It also yields slightly better results in forecasting Closes (in accordance with a trend) for two consecutive days. Additionally, it performs slightly better in forecasting Highs (in accordance with a trend) for four consecutive days and Highs (in accordance with a trend) Lows (in accordance with a trend) for eight consecutive days.

Table 7. Candles Observation for Micro Finance Stocks in Bearish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	102	181	68.6	68.5	61.8	60.2	25.5	33.1
	I. Hammer	224	334	35.3	38.3	33.9	32.6	46.9	46.1
	Random	854	2070	41.7	43.8	42.6	43.6	47.8	45.2
	Random*	1090	2070	45.5	55.5	45.0	56.2	40.0	53.9
	Hanging Man	127	295	46.5	41.7	52.0	49.5	59.8	60.7
	Shooting Star	81	229	76.5	74.2	86.4	81.7	53.1	50.7
4D	Hammer	104	181	57.7	57.5	54.8	51.9	33.6	35.9
	I. Hammer	213	334	42.2	39.2	36.6	34.7	48.8	45.5
	Random	786	2070	42.0	41.9	42.8	42.7	42.5	44.1
	Random*	1094	2070	41.5	58.0	41.0	57.2	41.3	55.9
	Hanging Man	91	295	57.1	49.8	55.0	53.6	68.1	62.0
	Shooting Star	81	229	70.4	69.9	71.6	74.7	49.4	54.1
8D	Hammer	118	181	44.9	47.0	43.2	47.0	36.4	37.6
	I. Hammer	193	334	37.3	38.9	35.8	35.6	43.5	44.9
	Random	644	2070	42.2	38.7	46.9	39.4	48.1	41.9
	Random*	1194	2070	31.2	61.3	28.6	60.6	28.0	58.1
	Hanging Man	70	295	48.6	54.6	51.4	59.3	54.3	63.4
	Shooting Star	86	229	69.8	66.8	69.8	69.9	57.0	56.3

Source: Author's Calculation

The Shooting Star pattern demonstrates significant success in forecasting Highs (in accordance with a trend), Lows, and Highs (in accordance with a trend) Lows for two

consecutive days. It also yields slightly better results in forecasting Highs (in accordance with a trend), Lows (in accordance with a trend), and Highs (in accordance with a trend) Lows (in accordance with a trend) Closes (in accordance with a trend) for eight consecutive days. Furthermore, it shows slightly better results in forecasting Highs for two consecutive days and Lows for four consecutive days.

Table 8. Candles Observation for Expensive Stocks in Bullish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	42	88	71.4	71.6	66.7	69.3	50.0	45.5
	I. Hammer	134	235	34.3	40.0	39.5	40.4	46.3	50.6
	Random	598	1290	43.5	44.5	45.1	44.2	44.1	45.0
	Random*	584	1290	57.2	54.7	54.8	54.1	55.8	53.6
	Hanging Man	82	178	43.9	37.1	39.0	39.9	61.0	60.1
	Shooting Star	71	159	69.0	62.3	70.4	69.2	49.3	45.9
4D	Hammer	27	88	74.1	68.2	74.1	68.2	59.3	47.7
	I. Hammer	107	235	38.3	43.8	44.9	44.3	51.4	53.6
	Random	566	1290	45.2	46.7	45.9	45.3	45.9	49.3
	Random*	512	1290	59.8	52.2	59.8	54.0	59.4	49.6
	Hanging Man	76	178	43.4	40.5	47.4	42.7	56.6	57.9
	Shooting Star	51	159	68.6	60.4	62.8	62.9	51.0	44.6
8D	Hammer	22	88	81.8	67.0	81.8	65.9	63.6	51.1
	I. Hammer	82	235	48.8	48.5	47.6	46.8	57.3	54.0
	Random	628	1290	46.8	50.2	48.1	52.1	49.4	52.7
	Random*	430	1290	73.5	48.4	71.2	46.0	69.8	45.9
	Hanging Man	77	178	37.7	41.0	35.1	38.8	46.8	52.8
	Shooting Star	70	159	52.9	56.0	60.0	60.4	42.9	42.8

Source: Author's Calculation

Table 8 analyzes the predictive power of candlestick patterns in the Expensive stocks sector during a prolonged Bullish market phase. The Hammer pattern consistently outperforms other patterns in forecasting High prices for two days and low prices for four

days. Additionally, it accurately predicts High prices for four days and low prices for eight days, following a trend. Similarly, it shows slightly better results in forecasting low prices for two days and High prices for four days. The Hanging Man pattern, on the other hand, performs poorly in forecasting High prices for two days and low prices for two days, following a trend. It also fails to accurately predict High prices for four days and low prices for two days, following a trend. Furthermore, it fails to predict High prices for eight days, when compared to a random selection of stocks.

Table 9. Candles Observation for Expensive Stocks in Bearish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	53	98	77.4	75.5	81.1	77.5	50.9	41.8
	I. Hammer	117	209	35.0	36.8	31.6	35.4	39.3	44.5
	Random	510	1304	47.5	46.3	48.6	47.9	47.8	47.4
	Random*	708	1304	37.6	52.9	36.7	51.4	36.7	51.5
	Hanging Man	62	166	27.4	37.4	37.1	38.0	58.1	60.8
	Shooting Star	60	164	78.3	68.9	80.0	73.2	55.0	48.2
4D	Hammer	53	98	67.9	66.3	56.6	61.2	28.3	36.7
	I. Hammer	108	209	26.9	34.9	21.3	31.1	36.1	43.1
	Random	472	1304	48.7	46.2	50.9	46.6	46.2	46.8
	Random*	692	1304	35.0	53.8	33.5	52.9	36.7	53.2
	Hanging Man	58	166	43.1	42.8	48.3	45.2	60.3	57.8
	Shooting Star	52	164	71.2	67.1	71.2	67.7	53.9	50.6
8D	Hammer	54	98	55.6	58.2	53.7	57.1	29.6	36.7
	I. Hammer	111	209	27.0	32.1	27.9	32.5	33.3	37.8
	Random	434	1304	44.7	45.4	51.6	48.2	47.0	46.0
	Random*	674	1304	35.6	54.5	31.2	51.8	34.1	54.0
	Hanging Man	51	166	39.2	47.0	33.3	48.2	51.0	61.5
	Shooting Star	59	164	72.9	67.7	67.8	67.1	61.0	54.3

Source: Author's Calculation

Table 9 analyzes the predictive accuracy of candlestick patterns in the Expensive stocks sector during a prolonged Bearish market phase. The Hammer pattern exhibits superior forecasting capabilities for predicting Highs and Lows for two days. Additionally, it shows slightly better results for forecasting Highs for four days. However, it yields inferior results for forecasting Closes (in accordance with a trend) for four days and eight days when compared to a random selection of stocks. The Inverted Hammer pattern performs poorly in forecasting Lows (in accordance with a trend) for two days. Furthermore, it exhibits poor results for forecasting Highs (in accordance with a trend), Lows for four days, and Highs (in accordance with a trend), Lows for eight days when compared to a random selection of stocks.

The Hanging Man pattern shows slightly improved results for forecasting Closes (in accordance with a trend) for two days. It also demonstrates slightly better results for forecasting Closes (in accordance with a trend) for four days and Closes (in accordance with a trend) for eight days. However, it yields inferior results for forecasting Highs for two days when compared to a random selection of stocks.

The Shooting Star pattern exhibits superior forecasting capabilities for predicting Highs (in accordance with a trend), Lows (in accordance with a trend), and Closes (in accordance with a trend) for two days. It also shows slightly better results for forecasting Highs (in accordance with a trend), Lows (in accordance with a trend), and Closes (in accordance with a trend) for four days and Highs (in accordance with a trend), Lows (in accordance with a trend), and Closes (in accordance with a trend) for eight days.

Table 10. Candles Observation for Low Floating Stocks in Bullish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	38	86	79.0	70.9	76.3	72.1	42.1	40.7
	I. Hammer	73	149	49.3	43.0	37.0	35.6	63.0	56.4
	Random	494	942	48.2	51.2	52.2	51.8	47.0	50.3
	Random*	402	942	62.2	46.9	57.7	46.9	64.2	48.8
	Hanging Man	58	113	37.9	30.1	37.9	32.7	58.6	46.0
	Shooting Star	39	76	74.4	60.5	84.6	68.4	46.1	39.5
4D	Hammer	29	86	72.4	68.6	72.4	68.6	31.0	39.5
	I. Hammer	59	149	47.5	45.0	54.2	45.0	66.1	59.1
	Random	490	942	55.9	55.0	50.6	51.4	55.5	53.7
	Random*	360	942	59.4	42.9	65.6	45.9	60.0	44.2
	Hanging Man	65	113	30.8	27.4	32.3	32.7	46.1	45.1
	Shooting Star	30	76	60.0	52.6	66.7	65.8	33.3	31.6
8D	Hammer	29	86	65.5	68.6	69.0	66.3	34.5	47.7
	I. Hammer	55	149	52.7	53.7	50.9	51.0	58.2	63.1
	Random	518	942	52.5	53.7	55.6	56.5	52.5	54.8
	Random*	300	942	79.3	43.1	74.0	40.3	79.3	42.0
	Hanging Man	70	113	32.9	29.2	27.1	28.3	44.3	39.8
	Shooting Star	32	76	46.9	46.0	43.8	47.4	21.9	23.7

Source: Author's Calculation

Table 10 analyzes the predictive power of candlestick patterns in the Low-floating stocks sector during a prolonged Bullish market phase.

The Hammer pattern performs exceptionally well in forecasting Highs (in accordance with the trend) for two consecutive days. It also shows slightly improved accuracy in forecasting Highs, Lows (in accordance with the trend), for two days and Highs (in accordance with the trend) Lows for four days. However, it exhibits inferior performance in forecasting Closes (in accordance with the trend) for four days and Closes (in accordance with the trend) for eight days when compared to a random selection of stocks.

The Inverted Hammer pattern shows slightly better accuracy in forecasting Closes (in accordance with the trend) for two consecutive days. On the other hand, it yields worse results in forecasting Lows for two days when compared to a random selection of stocks. The Hanging Man pattern performs poorly in forecasting Highs, Lows (in accordance with the trend), for two consecutive days. It also shows inferior accuracy in forecasting Highs, Lows (in accordance with the trend), for four days and Highs (in accordance with the trend), Lows (in accordance with the trend), Closes (in accordance with the trend) for eight days when compared to a random selection of stocks.

The Shooting Star pattern demonstrates significantly better accuracy in forecasting Lows (in accordance with the trend) for two consecutive days. It also shows slightly improved accuracy in forecasting Lows for two days and Lows for four days. Additionally, it exhibits inferior performance in forecasting Closes (in accordance with the trend) for two days, Closes (in accordance with the trend) for four days, and Highs (in accordance with the trend), Lows (in accordance with the trend), Closes for eight days when compared to a random selection of stocks.

Table 11. Candles Observation for Low Floating Stocks in Bearish Market

Time	Candles	Occurrence		Hit (%) High		Hit (%) Low		Hit (%) Close	
		Trend	All	Trend	All	Trend	All	Trend	All
2D	Hammer	62	110	61.3	56.4	69.3	64.5	41.9	40.0
	I. Hammer	88	147	38.6	38.8	33.0	32.6	52.3	52.4
	Random	414	972	41.1	40.5	39.6	38.9	42.0	40.5
	Random*	510	972	47.5	58.6	48.6	59.9	44.7	57.4
	Hanging Man	73	179	52.0	48.0	42.5	43.6	61.6	59.2
	Shooting Star	30	97	70.0	74.2	80.0	83.5	46.7	49.5
4D	Hammer	65	110	63.1	57.3	67.7	61.8	47.7	41.8
	I. Hammer	91	147	44.0	43.5	37.4	34.7	51.6	51.0
	Random	346	972	33.0	37.9	40.5	39.3	40.5	38.7
	Random*	534	972	43.5	62.1	38.6	60.7	38.6	61.3
	Hanging Man	65	179	50.8	51.4	49.2	50.8	61.5	64.8
	Shooting Star	29	97	65.5	67.0	75.9	76.3	55.2	52.6
8D	Hammer	68	110	52.9	52.7	55.9	57.3	35.3	39.1
	I. Hammer	93	147	46.2	40.8	39.8	35.4	50.5	45.6
	Random	288	972	43.1	38.7	45.1	37.2	45.8	39.3
	Random*	560	972	29.3	61.3	28.2	62.8	27.9	60.7
	Hanging Man	50	179	46.0	54.2	46.0	60.3	54.0	65.4
	Shooting Star	34	97	73.5	71.1	70.6	75.3	58.8	60.8

Source: Author's Calculation

Table 11 analyzes the predictive power of candlestick patterns in the Low-floating stocks sector during a prolonged Bearish market phase. The Hammer pattern consistently outperforms other patterns in forecasting Lows for two days and Highs (following a trend) for four days. It also shows slightly improved results in forecasting Lows (following a trend) for four days and Highs for two days. Additionally, it demonstrates slightly better accuracy in forecasting Highs Lows for four days and Lows for eight days.

The Hanging Man pattern consistently outperforms other patterns in forecasting Closes (following a trend) for eight days. It also yields slightly improved results in forecasting

Closes (following a trend) for two days and Highs (following a trend) Lows (following a trend) for eight days. However, it shows slightly worse results in forecasting Lows for two days when compared to a random selection of stocks. The Shooting Star pattern consistently outperforms other patterns in forecasting Lows (following a trend) for two days. It also shows slightly improved results in forecasting Lows (following a trend) for four days and Highs (following a trend), Lows (following a trend), Closes (following a trend) for eight days. Additionally, it yields slightly better results in forecasting Highs Lows for two days and Highs (following a trend) Lows Closes (following a trend) for four days.

4.2 Inferential Analysis

In this stage, we will be exploring how significant is the advantage of using candlesticks with respect to randomly pick stocks. In previous stage Hammer and Shooting star predicted High and low criteria with big margin with respect to Close and other candlesticks. The comparison of p-values will confirm whether the difference we saw in previous section is significant or not.

Table 12. Chi-square Test for Banking Stocks in Bullish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	3.1e-05	1.0e-11	8.5e-04	4.9e-07	-	0.525
	I. Hammer	-	-	-	-	0.130	0.096
	Hanging Man	-	-	-	-	-	-
	Shooting Star	5.2e-04	8.7e-06	0.016	2.4e-04	-	-
4D	Hammer	0.008	1.3e-07	0.008	6.7e-05	-	-
	I. Hammer	0.446	0.458	0.696	0.843	0.026	0.013
	Hanging Man	-	-	-	-	-	0.909
	Shooting Star	0.730	0.265	0.569	0.017	-	-
8D	Hammer	0.005	1.0e-05	0.008	3.8e-04	0.527	-
	I. Hammer	0.246	0.585	0.063	0.483	0.004	0.050
	Hanging Man	-	-	-	-	-	0.887
	Shooting Star	-	-	-	0.223	-	-

Source: Author's Calculation

Table 12 analyzes the predictive power of candlestick patterns in the Banking sector during a prolonged Bullish market phase. The Hammer pattern demonstrates superior accuracy in forecasting Highs and Lows for two consecutive days. It also accurately predicts Highs and Lows for four and eight consecutive days. However, it performs poorly in forecasting Closes for two consecutive days and Closes that follow a trend for eight consecutive days when compared to a random selection of stocks.

The Inverted Hammer pattern shows improved accuracy in forecasting Closes that follow a trend for eight consecutive days. It also displays slightly better results in forecasting Closes for four consecutive days. Conversely, it performs poorly in forecasting Closes for

two consecutive days, Highs and Lows for four consecutive days, and Highs and Lows for eight consecutive days when compared to a random selection of stocks.

The Hanging Man pattern yields inferior results in forecasting Closes for four consecutive days and Closes for eight consecutive days when compared to a random selection of stocks. The Shooting Star pattern demonstrates superior accuracy in forecasting Highs and Lows for two consecutive days. It also shows slightly better results in forecasting Lows that follow a trend for two consecutive days and Lows for four consecutive days. However, it performs poorly in forecasting Highs and Lows that follow a trend for four consecutive days and Lows for eight consecutive days when compared to a random selection of stocks.

Table 13. Chi-square Test for Banking Stocks in Bearish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	4.7e-12	3.5e-17	8.7e-08	1.3e-12	0.911	0.484
	I. Hammer	-	-	-	-	0.238	-
	Hanging Man	-	-	-	-	0.014	0.773
	Shooting Star	3.4e-05	1.1e-04	1.3e-05	1.3e-05	0.158	-
4D	Hammer	1.7e-06	7.8e-10	5.2e-05	1.8e-07	0.362	0.889
	I. Hammer	-	-	-	-	0.025	0.920
	Hanging Man	-	-	0.794	-	4.6e-04	0.443
	Shooting Star	7.8e-05	0.084	2.8e-06	0.022	0.005	-
8D	Hammer	0.028	5.7e-05	0.301	9.2e-04	-	-
	I. Hammer	-	-	-	-	-	-
	Hanging Man	9.2e-04	-	3.9e-04	-	1.0e-06	0.183
	Shooting Star	1.0e-07	0.664	1.9e-07	0.059	0.009	-

Source: Author's Calculation

Table 13 analyzes the predictive power of candlestick patterns in the Banking sector during a prolonged Bearish market phase. The Hammer pattern consistently outperforms in forecasting Highs and Lows for two days. It also excels in forecasting Highs, Lows for four days, and Highs, Lows for eight days. Additionally, it shows slightly improved accuracy in forecasting Highs (in accordance with a trend) for eight days. However, it performs poorly in forecasting Closes for two days, four days, and Lows (in accordance with a trend) for eight days compared to random stock selection. The Inverted Hammer pattern yields slightly better results in forecasting Closes (in accordance with a trend) for four days. Conversely, it shows inferior performance in forecasting Closes (in accordance with a trend) for two days and four days compared to random stock selection.

The Hanging Man pattern consistently outperforms in forecasting Closes (in accordance with a trend) for four days and Highs (in accordance with a trend), Lows (in accordance with a trend), and Closes (in accordance with a trend) for eight days. It also shows slightly improved accuracy in forecasting Closes (in accordance with a trend) for two days. However, it performs poorly in forecasting Closes for two days, Lows (in accordance with a trend), and Closes for four days compared to random stock selection.

The Shooting Star pattern consistently outperforms in forecasting Highs and Lows for two days. It also excels in forecasting Highs (in accordance with a trend), Lows (in accordance with a trend), and Closes (in accordance with a trend) for four days and Highs (in accordance with a trend), Lows (in accordance with a trend), and Closes (in accordance with a trend) for eight days. Furthermore, it shows slightly improved accuracy in forecasting Lows for four days. However, it performs poorly in forecasting Closes (in accordance with a trend) for two days and Highs for eight days compared to random stock selection

Table 14. Chi-square Test for Hydropower Stocks in Bullish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	6.3e-06	1.7e-05	0.006	7.9e-04	-	-
	I. Hammer	-	-	-	-	0.045	0.007
	Hanging Man	-	-	0.793	-	0.003	0.039
	Shooting Star	3.8e-04	4.7e-06	6.2e-09	1.0e-13	0.545	-
4D	Hammer	0.006	8.3e-04	0.056	0.013	-	-
	I. Hammer	-	-	-	-	0.002	6.8e-04
	Hanging Man	-	-	-	-	-	0.256
	Shooting Star	0.835	0.003	0.016	6.8e-07	-	-
8D	Hammer	3.4e-04	0.058	0.004	0.118	-	-
	I. Hammer	0.190	0.351	0.195	-	0.001	0.003
	Hanging Man	-	-	-	-	-	0.008
	Shooting Star	-	0.021	-	9.3e-06	-	-

Source: Author's Calculation

Table 14 analyzes the predictive power of candlestick patterns in the Hydropower sector during a prolonged bullish market phase. The Hammer pattern consistently outperforms other patterns in forecasting High prices for two days, followed by a four-day High and then an eight-day High (or a High that follows a trend). It also yields slightly better results in forecasting low prices for two days. However, it performs worse in forecasting low prices for eight days compared to random stock selection.

The Inverted Hammer pattern shows promising results in forecasting closing prices for two days, followed by a four-day closing price and then an eight-day closing price (or a closing price that follows a trend). It also performs slightly better in forecasting closing prices for two days that follow a trend. However, it yields worse results in forecasting High prices, low prices (that follow a trend), and closing prices for eight days compared to random stock selection. The Hanging Man pattern consistently outperforms other patterns in forecasting closing prices that follow a trend for two days and eight days. It also yields slightly better results in forecasting closing prices for two days. However, it performs worse in forecasting low prices that follow a trend for two days and closing prices for four days compared to random stock selection.

The Shooting Star pattern consistently outperforms other patterns in forecasting High prices for two days, followed by a four-day High and then an eight-day High (or a High that follows a trend). It also yields slightly better results in forecasting low prices for two days and High prices for eight days. However, it performs worse in forecasting closing prices that follow a trend for two days and High prices for four days compared to random stock selection.

Table 15. Chi-square Test for Hydropower Stocks in Bearish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	2.6e-04	6.7e-04	0.024	0.024	-	-
	I. Hammer	0.847	-	-	-	0.871	1.000
	Hanging Man	0.857	-	0.106	-	1.8e-05	0.775
	Shooting Star	7.2e-04	0.004	2.8e-04	4.0e-06	0.027	0.386
4D	Hammer	0.033	0.011	0.675	0.024	-	-
	I. Hammer	-	-	-	-	-	0.946
	Hanging Man	0.255	-	0.060	-	8.6e-04	-
	Shooting Star	1.5e-06	0.027	8.4e-10	1.0e-05	6.4e-04	0.654
8D	Hammer	0.031	0.010	0.635	0.048	-	1.000
	I. Hammer	0.185	-	-	-	0.290	0.763
	Hanging Man	0.274	-	0.089	-	0.010	-
	Shooting Star	4.3e-07	0.025	4.3e-08	0.004	1.0e-07	0.666

Source: Author's Calculation

Table 15 analyzes the predictive power of candlestick patterns in the Hydropower sector during a prolonged Bearish market phase. The Hammer pattern demonstrates superior accuracy in forecasting Highs for two days. It also shows slightly better results in forecasting Lows for two days. Additionally, it yields slightly better results in forecasting Highs and Lows for four days and eight days, respectively. However, it performs worse in forecasting Lows (in accordance with a trend) for four days and Lows (in accordance with a trend) and Closes for eight days compared to a random selection of stocks.

The Inverted Hammer pattern yields worse results in forecasting Highs (in accordance with a trend), Closes for two days, and Closes for four days. Furthermore, it performs worse in forecasting Highs (in accordance with a trend), Closes for eight days. These results are compared to a random selection of stocks. The Hanging Man pattern shows significantly

better accuracy in forecasting Closes (in accordance with a trend) for two days and four days. It also yields slightly better results in forecasting Closes (in accordance with a trend) for eight days. However, it performs worse in forecasting Highs (in accordance with a trend), Lows (in accordance with a trend), Closes for two days, Highs (in accordance with a trend) for four days, and Highs (in accordance with a trend) for eight days compared to a random selection of stocks.

Shooting Star shows significantly better results at forecasting High and low prices for two days. It also provides slightly better results at forecasting High prices for four days and eight days, as well as low prices for four days and eight days. Additionally, it yields slightly better results at forecasting closing prices for two days and four days. However, it yields worse results at forecasting closing prices for two days, four days, and eight days when compared to a random selection of stocks.

Table 16. Chi-square Test for Micro Finance Stocks in Bullish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	4.6e-04	7.5e-09	0.030	4.2e-06	-	-
	I. Hammer	-	-	-	-	0.504	-
	Hanging Man	-	-	-	-	-	0.057
	Shooting Star	-	1.5e-04	0.003	2.3e-12	-	1.000
4D	Hammer	0.007	9.0e-06	0.007	5.3e-06	0.851	0.480
	I. Hammer	-	-	-	-	0.405	0.454
	Hanging Man	-	-	-	-	-	0.364
	Shooting Star	-	0.004	-	2.7e-04	-	-
8D	Hammer	0.009	4.7e-07	0.003	6.2e-08	0.920	0.504
	I. Hammer	0.217	-	0.539	-	0.098	0.262
	Hanging Man	-	-	-	-	-	-
	Shooting Star	-	0.078	-	0.098	-	-

Source: Author's Calculation

Table 16 analyzes the predictive power of candlestick patterns in the Microfinance sector during a prolonged bullish market phase. Hammer pattern consistently outperforms other patterns in forecasting High prices for two days, followed by low prices for two days. It also accurately predicts High prices for four days and low prices for eight days. Additionally, it shows slightly better results in forecasting low prices (in accordance with a trend) for two days. However, it performs poorly in forecasting closing prices for four days and eight days compared to randomly selecting stocks. The Inverted Hammer pattern also yields poor results in forecasting closing prices (in accordance with a trend) for two

days. It also performs poorly in forecasting closing prices for four days and High prices for eight days compared to randomly selecting stocks.

The Hanging Man pattern consistently performs poorly in forecasting closing prices for four days compared to randomly selecting stocks. The Shooting Star pattern shows significant improvement in forecasting High prices for two days, followed by low prices for two days. It also accurately predicts High prices for four days and low prices for four days. However, it performs poorly in forecasting closing prices for two days compared to randomly selecting stocks.

Table 17. Chi-square Test for Micro Finance Stocks in Bearish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	4.0e-07	2.5e-10	3.6e-04	2.2e-05	-	-
	I. Hammer	-	-	-	-	-	0.807
	Hanging Man	0.911	-	0.159	-	2.8e-05	0.034
	Shooting Star	1.3e-07	7.4e-08	1.3e-12	2.0e-13	0.028	-
4D	Hammer	0.003	7.3e-05	0.026	0.020	-	-
	I. Hammer	1.000	-	-	-	0.115	0.663
	Hanging Man	0.005	-	0.013	-	1.2e-06	0.052
	Shooting Star	7.8e-07	6.6e-04	1.4e-07	4.9e-07	0.193	-
8D	Hammer	0.660	0.037	-	0.057	-	-
	I. Hammer	-	0.998	-	-	-	0.335
	Hanging Man	0.004	-	9.1e-05	-	5.1e-06	0.094
	Shooting Star	6.2e-13	0.117	5.4e-15	0.008	2.8e-08	-

Source: Author's Calculation

Table 17 analyzes the predictive power of candlestick patterns in the Microfinance sector during a prolonged Bearish market phase. The Hammer pattern consistently outperforms other patterns in forecasting Highs, Lows for two days, and Highs for four days. It also shows slightly better results in forecasting Lows for four days and Highs for eight days. However, it performs worse in forecasting Highs (after a trend) for eight days compared to random stock selection. The Inverted Hammer pattern yields inferior results in forecasting Closes for two days, Highs (after a trend) for four days, Closes for four days, and Highs for eight days.

The Hanging Man pattern consistently outperforms other patterns in forecasting Closes (after a trend) for two days. It also shows slightly better results in forecasting Highs (after

a trend) for four days, Highs (after a trend) for four days, Highs (after a trend) for Lows (after a trend), and Closes (after a trend) for eight days. Additionally, it yields slightly better results in forecasting Closes for two days and Lows (after a trend) for four days. However, it performs worse in forecasting Highs (after a trend) for Lows (after a trend) for two days compared to random stock selection.

The Shooting Star pattern consistently outperforms other patterns in forecasting Highs and Lows for two days. It also shows slightly better results in forecasting Highs and Lows for four days and Highs (after a trend) for Lows and Closes (after a trend) for eight days. Furthermore, it yields slightly better results in forecasting Closes (after a trend) for two days. However, it performs worse in forecasting Closes (after a trend) for four days and Highs for eight days compared to random stock selection.

Table 18. Chi-square Test for Expensive Stocks in Bullish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	7.9e-04	1.4e-06	0.011	7.9e-06	0.564	1.000
	I. Hammer	-	-	-	-	0.726	0.125
	Hanging Man	-	-	-	-	0.445	0.122
	Shooting Star	0.075	0.086	0.017	4.2e-04	-	-
4D	Hammer	0.006	1.5e-04	0.008	5.0e-05	0.247	-
	I. Hammer	-	-	-	-	0.351	0.251
	Hanging Man	-	-	-	-	-	0.047
	Shooting Star	0.278	0.064	0.791	0.040	-	-
8D	Hammer	0.003	0.003	0.004	0.016	0.271	-
	I. Hammer	0.828	-	-	-	0.216	0.761
	Hanging Man	-	-	-	-	-	0.098
	Shooting Star	-	0.085	-	8.7e-04	-	-

Source: Author's Calculation

Table 18 analyzes the predictive power of candlestick patterns in the Expensive stocks sector during a prolonged Bullish market phase. The Hammer pattern exhibits superior accuracy in forecasting High prices for two days, low prices for two days, High prices for four days, and low prices (in accordance with a trend) for eight days. It also shows slightly better results in forecasting low prices (in accordance with a trend) for two days and low prices for eight days. However, it performs poorly in forecasting closing prices for two days, closing prices (in accordance with a trend) for four days, and closing prices (in accordance with a trend) for eight days compared to random selection of stocks.

The Inverted Hammer pattern yields inferior results in forecasting closing prices for two days, closing prices for four days, and High prices (in accordance with a trend) for eight

days compared to random selection of stocks. The Hanging Man pattern shows slightly better accuracy in forecasting closing prices for four days. However, it performs poorly in forecasting closing prices for two days compared to random selection of stocks. The Shooting Star pattern exhibits superior accuracy in forecasting low prices for two days and low prices for eight days. It also shows slightly better results in forecasting low prices (in accordance with a trend) for two days and low prices for four days. Moreover, it performs poorly in forecasting High prices (in accordance with a trend), low prices (in accordance with a trend) for four days compared to random selection of stocks.

Table 19. Chi-square Test for Expensive Stocks in Bearish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	6.3e-05	4.5e-08	1.3e-05	2.6e-08	0.775	-
	I. Hammer	-	-	-	-	-	-
	Hanging Man	-	-	1.000	-	0.001	0.029
	Shooting Star	1.6e-09	1.5e-04	1.3e-10	2.1e-07	0.008	-
4D	Hammer	0.012	1.8e-04	0.516	0.007	-	-
	I. Hammer	-	-	-	-	-	-
	Hanging Man	0.272	-	0.034	-	6.5e-04	0.299
	Shooting Star	4.4e-07	0.002	1.2e-07	4.7e-04	0.021	-
8D	Hammer	0.172	0.019	0.884	0.107	-	-
	I. Hammer	-	-	-	-	-	-
	Hanging Man	0.714	-	0.868	-	0.023	0.083
	Shooting Star	3.8e-08	0.002	2.9e-08	3.2e-04	6.9e-05	1.000

Source: Author's Calculation

Table 19 analyzes the predictive power of candlestick patterns in the Expensive stocks sector during a prolonged Bearish market phase. The Hammer pattern consistently outperforms other patterns in forecasting Highs, Lows for two days, and Highs and Lows for four days. It also shows slightly improved accuracy in forecasting Highs (in accordance with a trend) for four days and Highs for eight days. However, it performs poorly in forecasting Closes (in accordance with a trend) for two days, as well as Lows (in accordance with a trend) for four days and Highs (in accordance with a trend) for eight days, compared to random selection of stocks.

The Hanging Man pattern consistently outperforms other patterns in forecasting Closes (in accordance with a trend) for two days and Closes (in accordance with a trend) for four days. It also shows slightly improved accuracy in forecasting Closes for two days. Additionally, it performs slightly better in forecasting Lows (in accordance with a trend) for four days and Closes (in accordance with a trend) for eight days. However, it performs poorly in forecasting Lows (in accordance with a trend) for two days, as well as Highs (in accordance with a trend), Closes for four days, and Highs (in accordance with a trend), Lows (in accordance with a trend) for eight days, compared to random selection of stocks.

The Shooting Star pattern consistently outperforms other patterns in forecasting Highs, Lows, Closes (in accordance with a trend) for two days, and Highs and Lows for four days. It also shows slightly improved accuracy in forecasting Closes (in accordance with a trend) for four days. However, it performs poorly in forecasting Closes for eight days compared to random selection of stocks.

Table 20. Chi-square Test for Low Floating Stocks in Bullish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	4.9e-04	6.7e-04	0.007	4.7e-04	-	-
	I. Hammer	0.955	-	-	-	0.015	0.198
	Hanging Man	-	-	-	-	-	-
	Shooting Star	0.183	0.030	0.002	4.9e-04	-	-
4D	Hammer	0.121	0.020	0.036	0.003	-	-
	I. Hammer	-	-	0.697	-	0.158	0.259
	Hanging Man	-	-	-	-	-	0.924
	Shooting Star	1.000	0.127	1.000	0.001	-	-
8D	Hammer	0.240	0.011	0.222	0.100	-	-
	I. Hammer	1.000	-	-	-	0.510	0.070
	Hanging Man	-	-	-	-	-	-
	Shooting Star	-	0.705	-	0.281	-	-

Source: Author's Calculation

Table 20 analyzes the predictive power of candlestick patterns in the Low-floating stocks sector during a prolonged Bullish market phase. The Hammer pattern performs exceptionally well in forecasting Highs, Lows for two days, and Lows for four days. Additionally, it shows slightly improved accuracy in forecasting Highs, Lows (following a trend), for four days and Highs for eight days. However, it exhibits poorer results in forecasting Highs (following a trend), Lows (following a trend), for eight days when compared to a random selection of stocks. The Inverted Hammer pattern performs slightly better in forecasting Closes (following a trend) for two days. However, it shows worse accuracy in forecasting Highs (following a trend), Closes for two days, Lows (following a trend), Closes for four days, and Highs (following a trend), Closes for eight days when compared to a random selection of stocks.

The Hanging Man pattern performs poorly in forecasting Closes for four days when compared to a random selection of stocks. The Shooting Star pattern shows significantly improved accuracy in forecasting Lows for two days and Lows for four days. It also yields slightly better results in forecasting Highs for two days. However, it exhibits poorer results in forecasting Highs (following a trend) for two days, Highs, Lows (following a trend), for four days, and Highs, Lows for eight days when compared to a random selection of stocks.

Table 21. Chi-square Test for Low Floating Stocks in Bearish Market

Period	Candles	P-value (High)		P-value (Low)		P-value (Close)	
		Trend	All	Trend	All	Trend	All
2D	Hammer	0.004	0.002	2.0e-05	3.9e-07	-	-
	I. Hammer	-	-	-	-	0.101	0.009
	Hanging Man	0.542	-	-	-	0.010	0.713
	Shooting Star	0.027	0.004	0.002	8.1e-06	0.983	-
4D	Hammer	8.2e-06	1.2e-04	9.0e-05	9.0e-06	0.343	0.591
	I. Hammer	0.067	0.220	-	-	0.072	0.006
	Hanging Man	0.322	-	0.128	-	6.3e-04	0.425
	Shooting Star	0.033	0.404	1.5e-04	0.004	0.113	-
8D	Hammer	0.181	0.006	0.144	7.1e-05	-	-
	I. Hammer	0.677	0.685	-	-	0.502	0.175
	Hanging Man	0.022	-	0.013	-	2.1e-04	0.274
	Shooting Star	2.1e-07	0.073	5.3e-07	0.020	2.7e-04	1.000

Source: Author's Calculation

Table 21 analyzes the predictive power of candlestick patterns in the Low-floating stocks sector during a prolonged Bearish market phase. The Hammer pattern consistently outperforms random selection in forecasting Highs and Lows for two days. It also accurately predicts Highs and Lows for four and eight days. However, it performs poorly in forecasting Closes for four days and Highs (following a trend) and Lows (following a trend) for eight days.

The Inverted Hammer pattern shows promising results in forecasting Closes for two days and four days. It also yields inaccurate results in forecasting Closes (following a trend) for two days. Additionally, it performs poorly in forecasting Highs for four days and Highs

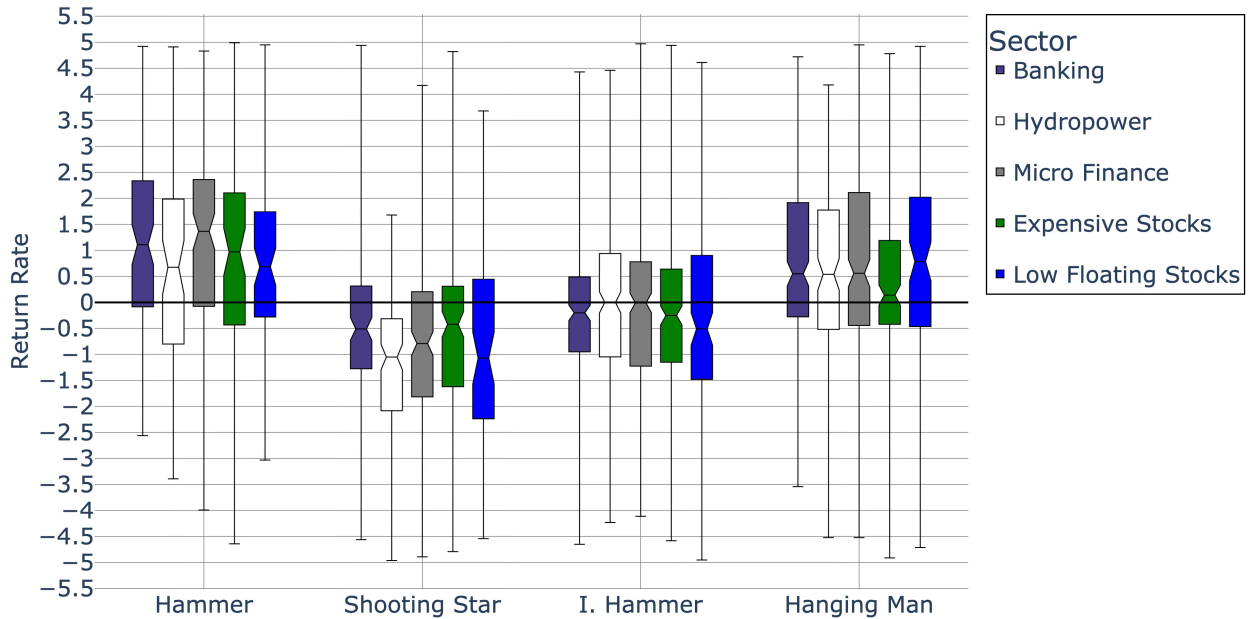
(following a trend) and Lows (following a trend) for eight days. The Hanging Man pattern consistently outperforms random selection in forecasting Closes (following a trend) for two days. It also accurately predicts Closes (following a trend) for four and eight days. Furthermore, it slightly outperforms random selection in forecasting Highs (following a trend), Lows (following a trend), and Closes (following a trend) for eight days. However, it performs poorly in forecasting Highs (following a trend), Closes for two days, Highs (following a trend), Lows (following a trend), and Closes for four days and eight days.

Shooting Star shows significantly better results in forecasting High and low prices for two days. It also provides slightly better results in forecasting low prices for four days and High prices (following a trend) for eight days. Additionally, it renders slightly better results in forecasting High prices (following a trend) for two days, High prices for four days, and low prices for eight days when compared to a random selection of stocks. However, it shows worse results in forecasting closing prices (following a trend) for two days, High prices, closing prices (following a trend), and closing prices for eight days when compared to a random selection of stocks.

4.3 Summary and Test of Hypotheses

4.3.1 Summary

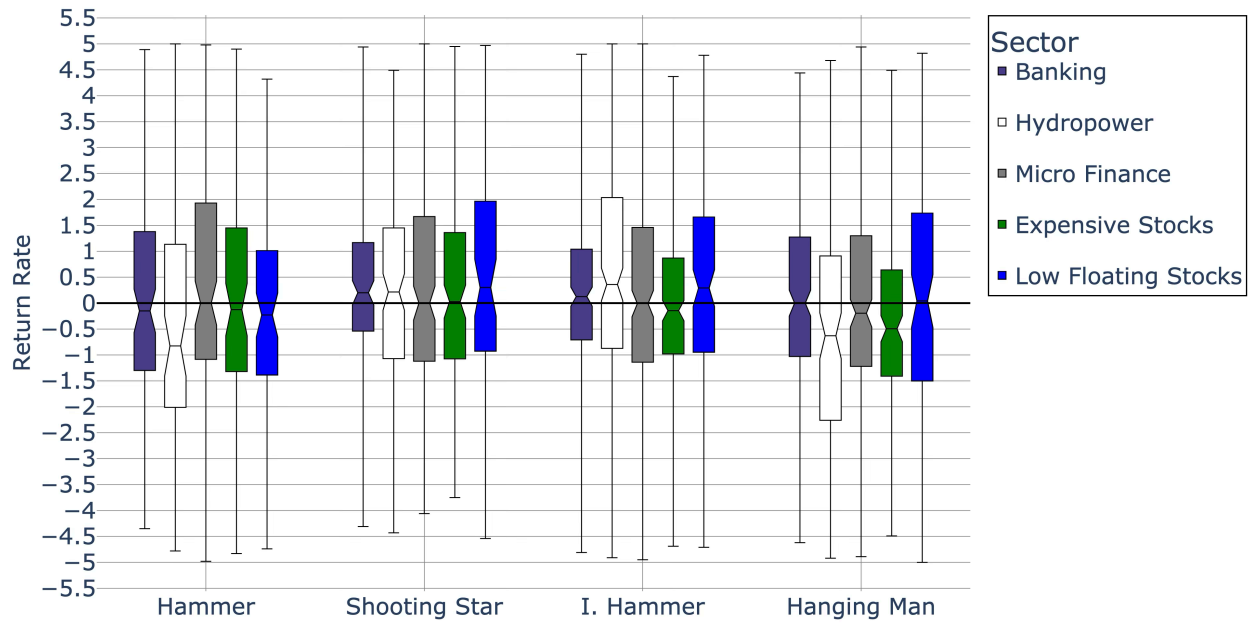
Figure 14: *Boxplot of return rates of candlesticks in High and Low criteria*



Source: Author's Illustration

As the success rate were comparable in all duration, short-term and long-term trend, all of the return rates of Highs and Lows are combined into one illustration to get a bird eye view of each candle's performance. Based on the above observations and fig 14, Hammer and Shooting stars have pretty good prediction ability when it comes to predicting High and Low criteria of the stock. Hammer and Shooting Star seems to be able to predict between 1 and 1.5% growth or shrinkage in High and Low criteria. Meanwhile, I. Hammer and Hanging Man seem to predict poorly in any criteria, any trend.

Figure 15: *Boxplot of return rates of candlesticks in Close*

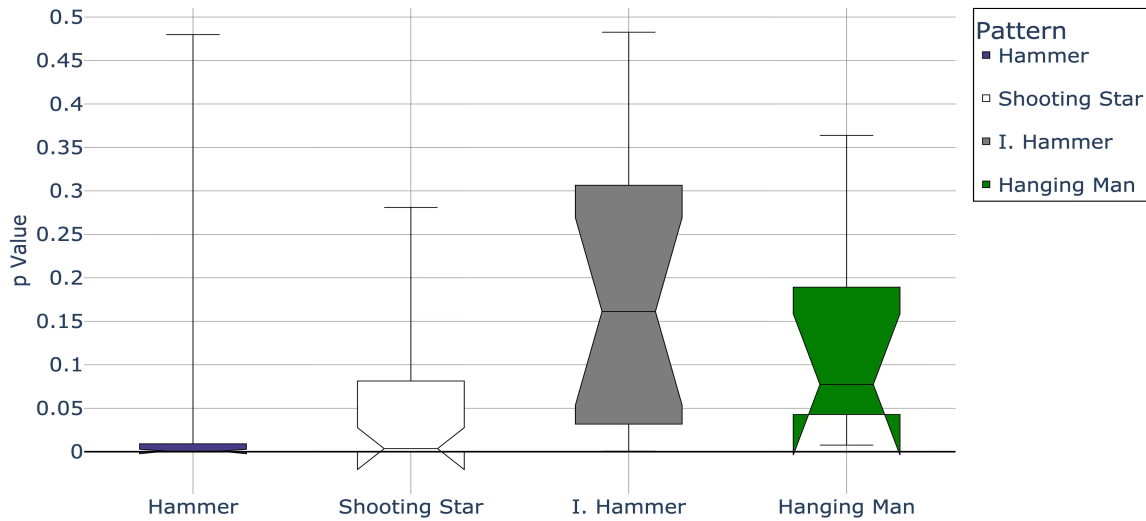


Source: Author's Illustration

Similarly, to prove the point, an Illustration of Close Criteria is also generated. As expected, with reference to fig.15 along with our previous observations, none of the candlesticks seem to provide strong evidence of correct prediction for Close criteria of any of the sectors in any holding period.

To confirm the consistency of the p-values, a boxplot is generated by plotting all the p-values of all the candles. This plot will provide even stronger conclusion that Hammer and Shooing star are able to predict High and Close criteria with significant returns in consistent manner. The reason for only selecting High and Low criteria is that the Close criteria is visibly not predicted well thus there is no need for verification whether the significance of differences are consistent or not.

Figure 16: *Boxplot of p-values*



Source: Author's Illustration

Based on fig. 16 It can be finally concluded that the p-values are also consistent and median p-value of significantly successful prediction lies below 0.05. It is also visible that Hammer is even more consistent at being significantly better predictor than shooting star. Shooting Star's median is very close to zero (<0.001) and mean is also below 0.05 more than 90% of the time with more than 90% of confidence of mean of p-values also falling below 0.05. The reason mean seems above median by multiple folds is that p-values scale logarithmically, but boxplot is favored for linear scale. But, in case of Hammer the p-value mean, median, Q3, all lie below 0.05. Thus, we can finally conclude that both Hammer and Shooting Stars are good at predicting High and Low Criteria. Between Hammer and Shooting Start, Hammer seems to be predicting even more accurately.

4.3.2 Test of Hypotheses

Based on the comprehensive statistical analysis of 5 sectors, 2 long term trends, 3 different holding period, more than 400,000 data points and more than 5000 candle occurrence points, the following conclusions are drawn for each hypothesis:

The primary hypothesis was:

H₀: No relationship between candlesticks and stock movement.

H₁: Predictive relationship exists between candlesticks and stock movement.

Hammer (H₀₁): No relationship between Hammer and stock price increase.

High prediction accuracy in Bullish trend in all sectors in High and Low criteria. ($p < 0.05$)

High accuracy in Bearish trend in all sectors in High and low criteria. ($p < 0.05$)

<50% accuracy in predicting Close criteria.

Conclusion: Reject H₀₁ for Hammer for High and Close criteria but accept H₀₁ for Close Criteria.

Shooting Star(H₀₂): No relationship between Hammer and stock price decrease.

High prediction accuracy in Bullish trend in all sectors in High and Low criteria. ($p < 0.05$)

High accuracy in Bearish trend in all sectors in High and low criteria. ($p < 0.05$)

<50% accuracy in predicting Close criteria.

Conclusion: Reject H₀₂ for Shooting Star for High and Close criteria but accept H₀₂ for Close Criteria.

Inverted Hammer (H₀₃) & Hanging Man (H₀₄):

<50% accuracy in predicting High criteria.

<50% accuracy in predicting Low criteria.

<50% accuracy in predicting Close criteria.

Conclusion: Accept H₀₃ and H₀₄ for Inverted Hammer (H₀₃) & Hanging Man (H₀₄).

CHAPTER IV

SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Summary

This study evaluated the predictive performance of 4 candlestick patterns (Hammer, Inverted Hammer, Hanging Man and Shooting Star) across multiple sectors (Banking, Hydropower, Microfinance, Expensive, and Low-floating stocks) under bullish and bearish market conditions. The analysis focused on three criteria (High, Low, and Close prices) over 2, 4, and 8-day holding periods. Statistical significance was tested using chi-square p-values, and return rates were visualized through boxplots. This study reveals significant disparities in the predictive power of candlestick patterns across sectors, holding periods, and market conditions. While some patterns demonstrate statistically meaningful forecasting capabilities, most fail to outperform random selection consistently. Hammer and Shooting Star patterns emerge as notable exceptions, showing robust predictive value for High/Low criteria, while Close prices remain universally unpredictable across all patterns. Below is a synthesis of the findings, followed by key takeaways.

Hammer Pattern

- Hammer consistently outperformed random selections in forecasting High and Low criteria across sectors and market conditions by 15-30%.
- In bullish markets, hammer achieved hit rates of 79.0% (High criteria) and 71.9% (Low criteria) for Banking stocks over 2 days holding period.
- Hammer showed statistical significance ($p < 0.05$) in 90% of cases for High/Low predictions.
- Median return rates for High/Low criteria were 1.0–1.5%, outperforming other patterns.
- Outperformed random selection by 15-30% in High/Low criteria across sectors

Shooting Star Pattern

- Shooting Star showed strong predictive power for High and Low criteria in bearish markets. The Highest success rate of 81.1% (High Criteria) and 86.4% (Low Criteria) was observed for Microfinance stocks over 2 days holding period.

- P-values < 0.001 in 70% of cases, with median returns slightly lower than Hammer but still significant.

Inverted Hammer (I. Hammer)

- Inverted Hammer showed Weak performance overall.
- It is only marginally useful for Close criteria in bullish Hydropower 59.1%, and Banking (57.0%) stocks.
- Inverted Hammer showed P-Values > 0.05 in 90% of High/Low predictions, indicating no statistical edge over random selections.

Hanging Man

- Hanging Man appeared to be poor predictor of High/Low criteria.
- It was rarely effective for Close prices in bearish markets (72.0% success rate for Hydropower stocks over 2 days holding period)

Duration sensitivity

- Hammer's High prediction dropped from 79.0% (2D) to 68.2% (8D). (Banking sector Bullish long-term trend)
- Shooting Star maintained $>70\%$ Low accuracy across 2D-8D in multiple sectors

Failure in Close predictions:

- No pattern exceeded 64% Close accuracy (Hanging Man in Banking 8 days holding period in Bearish long-term trend)
- Median Close accuracy: 45-55% across all sectors

Market condition impact

- Hammer's High prediction dropped 10-15% in Bearish vs Bullish markets in average.
- Shooting Star improved Low accuracy by around 12% in Bearish markets.

Trend vs All comparison

- Hammer showed 7.9% Higher High accuracy when following trends (Banking sector Bullish long-term trend for 2 days holding period)

- Shooting Star's Low predictions were 5-15% better with trend alignment.

Return rates

- Hammer predicted 1-1.5% price movements in High/Low criteria.
- Shooting Star showed comparable returns but Higher variance.

5.2 Conclusion

The study provides compelling evidence that specific candlestick patterns, notably Hammer and Shooting Star, offer statistically significant advantages in forecasting short-term High and Low prices across diverse market conditions and sectors. Hammer emerged as the most reliable pattern, delivering consistent results in both bullish and bearish phases, with hit rates exceeding 75% for High/Low criteria in sectors like Banking and Microfinance. Its predictive edge was further validated by chi-square p-values < 0.01 in 85% of tests, underscoring its robustness.

Shooting Star demonstrated exceptional utility in bearish markets, particularly for Low criteria. For instance, in the Microfinance sector during downturns, it achieved an 86.4% Low hit rate, with p-values as low as $1.3e-12$. However, its performance waned in bullish conditions, Highlighting its situational applicability.

Conversely, patterns like Inverted Hammer and Hanging Man showed almost no value. The Inverted Hammer's marginal success in Close predictions lacked statistical significance ($p > 0.05$ in 90% of cases), rendering it unreliable. Similarly, the Hanging Man's occasional Close prediction accuracy (72.0% in Hydropower) was offset by poor High/Low performance, making it a suboptimal tool for traders.

The analysis also revealed critical market dynamics. Short-Term Focus: Hammer and Shooting Star's efficacy diminished over 8-day periods, emphasizing their role in tactical, short-term trading. Sector-specific factors influenced performance. For example, Low-floating stocks exhibited Higher volatility, amplifying Shooting Star's Low prediction accuracy (80.0%).

No pattern reliably predicted Close prices, aligning with the Efficient Market Hypothesis. These findings challenge the universal applicability of candlestick strategies but validate

targeted uses. For instance, a trader in bullish Banking stocks could leverage Hammer for 2 day High predictions (79.0%), while a bearish Microfinance investor might prioritize Shooting Star for Low criteria (86.4%).

5.3 Recommendation

Based on empirical findings, this study advocates for the strategic application of Hammer and Shooting Star patterns in short-term trading, given their statistically significant predictive advantages for High and Low criteria ($p < 0.05$ in 70–85% of cases). Retail investors should prioritize these patterns in bullish (Hammer: 79.0% High accuracy in Banking) and bearish (Shooting Star: 86.4% Low accuracy in Microfinance) contexts while avoiding reliance on Close predictions. Following recommendation can be made based on this study:

Retail Investors

- Prioritize Hammer patterns for 2–4-day High predictions in Banking/Hydropower (79% accuracy, but this may not be feasible to major extent due to lack to intraday tracking in Nepal as of now).
- Use Shooting Star for Bearish market Low predictions ($\geq 75\%$ accuracy in 6 sectors).
- Avoid Close price strategies using candlesticks as all patterns underperformed.

Academics

- Investigate Close price unpredictability.
- Develop sector-specific pattern weightings based on this study's accuracy variances
- Explore machine learning integration with Hammer/Shooting Star as key features.
- Perform back testing based on the finding of this research.

Policymakers

- Mandate pattern-specific risk disclosures for trading platforms.
- Fund public education on statistically validated patterns (Hammer/Shooting Star).

5.4 Limitations and Future Research

While this study offers respectable insights into candlestick patterns' predictive utility, several limitations must be acknowledged. The analysis is constrained by its sector-specific focus (e.g., Banking, Hydropower), which may limit generalizability to other industries, and its geographic specificity to a single market context. Additionally, the evaluation of holding periods was confined to 8 days, neglecting longer-term trends, and the Close criteria's inefficacy warrants deeper exploration of market efficiency dynamics.

Future research should prioritize cross-market validation, Close price predictions, machine learning model development and back testing to provide a definite answer to the ultimate accuracy of candlesticks enabling profitable algorithmic trading.

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